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THE EFFECT OF VARYING WEIGHTLOAD INTENSITIES ON BICEPS CURL TECHNIQUE

by

Jeffrey C. Gallhouse

A Thesis
Submitted to the
Faculty of The Graduate College
in partial fulfillment of the
requirements for the
Degree of Master of Arts
Department of Health, Physical Education, and Recreation

Western Michigan University
Kalamazoo, Michigan
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THE EFFECT OF VARYING WEIGHTLOAD INTENSITIES ON BICEPS CURL TECHNIQUE

Jeffrey C. Gailhouse, M.A.

Western Michigan University, 1992

This study described biomechanical deviations from proper biceps curl technique specifically in the shoulder, trunk, body and knee angles, that occurred in response to varying intensities of an individual's 1 RM (repetition maximum). Ten college age males participated in the study. Subjects were required to perform one repetition at 60%, 70%, 80%, 90% and 100% of his 1 RM. The only differentiating factor was the order in which the five intensities were performed. Condition order was randomized to minimize bias. Movement was filmed with a high-speed motion picture camera, digitized and then analyzed with computer software.

The findings indicated that a relationship existed between resistance and the magnitude of movement in the selected angles. As resistance increased, angular movement correspondingly increased.

It was concluded that training at intensities less than 90% of an individual's 1 RM was more conducive to proper technique and would still allow for optimal strength gains.

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Jeffrey C. Gallhouse

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TABLE OF CONTENTS

ACKNOWLEDGEMENTS ii

LIST OF TABLES vi

LIST OF FIGURES vii

CHAPTER

I. INTRODUCTION 1

Statement of the Problem 1

Purpose of the Study 1

Delimitations 2

Limitations 2

Assumptions 2

Definition of Terms 3

II. REVIEW OF LITERATURE 5

Biomechanical Implications 6

Mechanics--Levers 6

Center of Gravity 7

Strength Training Guidelines 7

Principles 7

Program Variables 8

Strength Training Programs 9

Technique Implications 12

Proper Technique 12

Training Related Injuries 13

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Table of Contents--Continued

CHAPTER

Biceps Curl Technique 14

III. DESIGN AND METHODOLOGY 16

Subject Selection 16

Instrumentation 17

Photo-Sonic 1PL Camera 17

Vanguard Motion Analyzer 17

Peripheral Instrumentation 17

Experimental Procedures 18

Data Analysis 19

Statistical Procedures 21

IV. RESULTS AND DISCUSSION 22

Results. 22

Homogeneity of Variance 23

Shoulder Angle 24

Trunk Angle 29

Body Angle 34

Knee Angle 38

Discussion 39

Shoulder Angle 39

Trunk Angle 41

Body Angle 42

Knee Angle 43

Related Factors 44

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Table of Contents--Continued

CHAPTER

V. SUMMARY, FINDINGS, CONCLUSIONS, AND RECOMMENDATIONS 45

Summary 45

Findings 46

Conclusions 46

Recommendations 47

APPENDICES

A. Human Subjects Review Board Acceptance Form 49

B. Informed Consent 51

C. Subject-Treatment Order 53

D. Anatomical Landmarks for Joint Angle Calculation 55

E. Subject Raw Angle Data 57

F. Group Means by Joint Angle 68

BIBLIOGRAPHY 71

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LIST OF TABLES

1. Summary Table for Homogeneity of Variance 25

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LIST OF FIGURES

1. Subject 7 Shoulder Angle Movement Pattern 26
2. Subject 2 Shoulder Angle Movement Pattern 27
3. Subject 3 Shoulder Angle Movement Pattern 28
4. Subject 2 Trunk Angle Movement Pattern 30
5. Subject 9 Trunk Angle Movement Pattern 31
6. Subject 7 Trunk Angle Movement Pattern 32
7. Subject 5 Trunk Angle Movement Pattern 33
8. Subject 7 Body Angle Movement Pattern 35
9. Subject 6 Body Angle Movement Pattern 36
10. Subject 1 Body Angle Movement Pattern 37
11. Subject 9 Body Angle Movement Pattern 38

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CHAPTER I

INTRODUCTION

A great deal of confusion and dissension exists among professionals in the field of sport pertaining to weight training principles. This dilemma is quite evident in the area of strength training. Strength trainers are divided over the issue of training intensities. Some trainers maintain that higher training intensities, at the expense of proper technique and form, produce optimal gains in terms of strength development. Others, however, feel that training at a slightly lower intensity level emphasizing proper technique will yield optimal results. Research to determine the effects that various training intensities have on proper technique is for the most part, nonexistent. Even though strength gains are to be expected with most training intensities, an optimal training zone must be established through research to maximize the potential benefits of resistance training.

Statement of the Problem

The problem in this study was to determine, in the sagittal plane, deviations that occur as a result of varying intensities in relation to the one repetition maximum (1 RM). More specifically, varying intensities may elicit a variety of biomechanical movements that are incorrect with regard to proper technique.

Purpose of the Study

With naked eye observation it can be ascertained that high intensity resistances cause a greater degree of technique deviation than lighter intensities. For

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example, an individual performing a biceps curl at 90% of 1 RM will demonstrate a marked difference in technique when compared to a curl at 60% of 1 RM. Studies to determine the magnitude of differences in terms of biomechanical technique among different training intensities were not found in the literature. In order to obtain the most benefit from a strength training program, an individual must train at optimal intensity levels. Studies investigating the biomechanical changes that occur during the execution of the biceps curl need to be conducted to facilitate the establishment of optimal training intensities.

Delimitations

The study was delimited to the following characteristics:

1. Participants were between the ages of 19 and 23 years.
2. Subjects were males currently enrolled at Western Michigan University.
3. None of the subjects received instruction about maintaining proper technique prior to testing.
4. Movement was only filmed in the sagittal plane.

Limitations

A limitation of this investigation was that subjects were randomly selected from respondents to verbal announcements made in general physical education weight training classes. Consequently, most of the subjects had prior weight training experience. The presence of these characteristics may limit the application of results to other populations.

Assumptions

The study was conducted under the following assumptions:

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1. The participants performed consistently with maximal effort on all trials so that true anatomical mechanics could be recorded.

2. The instrumentation used for data collection and analysis was accurate and reliable.

Definition of Terms

The following terms and definitions are pertinent to the understanding of this study:

1. Antagonistic muscle--the muscle that produces the opposite joint action to that of the prime mover muscle.

2. Concentric contraction--muscular contraction in which the muscle exerts force and shortens.

3. Eccentric contraction--muscular contraction in which the muscle exerts force and lengthens.

4. Intensity--the percentage of weight lifted in relation to the 1 RM.

5. Joint extension--increasing the angle between two adjacent segments.

6. Joint flexion--decreasing the angle between two adjacent segments.

7. One repetition maximum (1 RM)--the maximum amount of weight lifted one time with correct form for a specific exercise.

8. Prime mover muscle--the muscle that contracts to accomplish the desired movement.

9. Range of motion (ROM)--for the biceps curl a movement from a position of full extension to a position of full flexion.

10. Repetition--moving a weightload from a position of full extension to a position of full flexion and back to the initial position of full extension.

11. Treatment--the specific resistance at which each subject was tested.

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12. Trunk inclination--movement of the upper body torso, either forward or backward, from a normally erect position.

13. Weightload--the amount of resistance, expressed in pounds, with which an individual exercises.

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CHAPTER II

REVIEW OF LITERATURE

The biceps brachii and brachialis are the elbow flexors. When the upper arms and elbows remain in a fixed position, the elbow flexors serve as the prime mover muscle group which results in a more beneficial training effect. For this reason, the biceps curl must be performed with very strict technique. Very few, if any, studies have been conducted concerning the biomechanical changes that occur as the biceps curl is executed. Additionally, no research has been done on the effects that varying weightload intensities have on biceps curl technique. In order to accurately analyze an individual's lifting technique and to establish programs implementing optimal training intensities with proper technique, an understanding of the fundamentals of strength training is necessary. Knowledge of weight training principles, variables, and programs is essential. Equally important is an understanding of how proper technique influences the outcomes of a weight training program and aids in injury prevention. The implications that the field of biomechanics has on strength training also merit consideration. Thus, an accurate biomechanical analysis of the biceps curl can only be made with background knowledge in the fields of strength training and biomechanics. This chapter is organized into three sections: (1) biomechanical implications, (2) strength training guidelines, and (3) technique implications.

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Biomechanical Implications

Mechanics--Levers

The effectiveness that weight training has on improving strength is influenced by the anatomical arrangements of limbs, joints and muscles. An understanding of how these body parts interact to form a system of levers helps strength coaches design more effective training programs. A lever system is comprised of three components: a fulcrum (joint), a resistance (weight), and a force (muscle contraction). Within the body there are three classes of levers. Each class of lever serves a specific purpose in the body. In a Class I lever system the fulcrum is located between the resistance and the force. Class I lever systems are responsible for maintaining equilibrium. A teeter-totter is an example of a Class I lever. Individuals on each end of the teeter-totter serve as the resistance and the force, both acting downward, with the fulcrum located between them. The resistance is located between the fulcrum and the force in a Class II lever system. This type of lever system serves to conserve energy by reducing the amount of force necessary to accomplish the desired movement. An example of a Class II lever system is a wheel barrow. The axle serves as the fulcrum, the weight in the bucket serves as the resistance, and the force is applied by the individual lifting up on the handles. In a Class III lever system the force is situated between the fulcrum and the resistance. This lever system is responsible for speed and range of movement. Performing the biceps curl is an example of a Class III lever system. The force is the contraction of the biceps muscle, the elbow serves as the fulcrum, and the weight being held in the hand serves as the resistance. With increased resistance, the biceps must exert greater force in order to accomplish the desired movement throughout the range of motion.

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Center of Gravity

The amount of force a muscle must exert is influenced by its position in relation to the body's center of gravity. The weight of a body acts vertically downward through its center of gravity, establishing a line of action (Watkins, 1983). This can be illustrated by drawing a vertical line through the center of a body's profile view. This line of action is a key factor in terms of body stability. Watkins (1983) stated that a body is most stable when its line of action falls within its base of support. During the execution of the biceps curl the body's center of gravity changes in relation to the line of action and stability is maintained through body movement. When an individual performs a strength exercise against a correct resistance needless body movement will be minimized. A joint's position in relation to the body's line of action also influences the amount of force a muscle must exert to accomplish a desired movement. The greater the distance between a joint and the line of action, the greater the force required from the muscles involved with joint movement. When movement about the shoulder joint during the biceps curl is kept to a minimum the role of the elbow flexors as the prime movers of the elbow joint is emphasized. Training at an optimal intensity level, from a biomechanical perspective, will allow the individual to maintain proper technique and experience maximal strength gains.

Strength Training Guidelines

Principles

The goal of strength training programs is to elicit increases in muscular strength and/or muscular endurance. Muscular strength is the ability of a muscle to contract against a resistance. The ability of a muscle to make repeated contractions is muscular endurance. Muscular strength and muscular endurance are interrelated in

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that an increase in one component will generally result in an increase in the other component. Three basic principles must be implemented to derive maximum gains from strength training programs. First, muscular adaptations only occur when the muscles are overloaded--forced to contract against near maximum resistances. This principle has been termed the overload principle. Second, muscles adapt to the load they are subjected to, therefore it is necessary to continually overload the muscle in order to promote further strength gains. As strength increases from training, so must the amount of resistance that an individual exercises against. If the resistance is not increased, the existing strength level will be maintained, not improved. Therefore, the intensity in terms of the resistance must be progressively increased to promote further strength gains, a principle known as progressive resistance. Third, the principle of specificity implies that muscle adaptations are specific to the type of training that is performed. If a muscle is engaged in endurance type exercises, the endurance capacity of a muscle is increased. If a muscle is engaged in heavy resistance training, muscular strength is increased. The inclusion of these principles into a training program will allow the individual to maximize strength gains.

Program Variables

Strength training programs are organized into repetitions (the number of times that a weight is lifted) and sets (a sequence of repetitions). An optimal combination of sets and repetitions to achieve optimal strength gains has not been established. The most successful programs implement a range of 6 to 12 repetitions per set with a minimum of three sets. Strength programs are based upon the amount of weight that can be lifted for a selected number of repetitions before fatigue sets in. This resistance has been termed the repetitions maximum. Most strength training programs advocate exercising with a weight that can be lifted ten times in succession; the ten repetitions

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maximum, or 10 RM. Once the 10 RM can be performed for all sets, the weightload should be increased. As the resistance increases, strength correspondingly increases. The weightload and number of repetitions and sets within a program can be manipulated to meet the goals of the individual. For optimal gains in muscular strength, more sets with fewer repetitions are performed against a heavier resistance. For example, the 7 RM performed against 90% of the 1 Rm for five sets might be executed. Muscular endurance is improved when more repetitions are performed against a lighter resistance. A muscular endurance protocol might require 4 sets of the 10 RM to be performed against 70% of the 1 RM. Regardless of the desired outcomes of a strength training program, the resistance must be progressively increased as the individual is able to successfully complete the desired number of sets and repetitions so that the muscle is consistently overloaded.

Strength Training Programs

Three types of strength training programs are used to elicit strength gains; isometric, isokinetic, and isotonic. Each modality has advantages as well as limitations.

Isometric training involves a static muscle contraction in which the length of the muscle remains constant (does not shorten or lengthen) when force is exerted against a fixed resistance. Isometric exercises are usually performed against a stationary resistance such as a wall or a training partner. The results of a study that involved the execution of a daily maximum isometric contraction for a one second duration demonstrated a 5% weekly increase, over an eight week time period, in strength over initial strength levels (Hettinger & Muller, 1953). Despite the improvements in strength, isometric training has severe limitations. Progress, in terms of strength development, is difficult to evaluate (McArdle, Katch, & Katch, 1986). The absence of identifiable measures of resistance (weight) makes it difficult to chart improvement. Isometric

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exercises do not involve limb movement, therefore muscle activation and subsequent strength gains are specific to the joint angle at which contraction occurred (Lindl, 1979; Zabik, 1967). In order for isometric training to be overtly beneficial several angles throughout the range of motion must be exercised for each respective muscle group. The time-to-benefit ratio is less than desirable and therefore this form of training is frequently ignored.

Isokinetic strength training is an outgrowth of isotonic training. This form of training requires muscular contraction to occur at a constant velocity (Powers & Howley, 1990). As the length of the muscle changes, the resistance varies proportionally to match the force exerted by the muscle. Isokinetic machines allow the exerciser to exert maximal force throughout the entire range of motion. This is accomplished by combining variable resistance capabilities with a speed governing feature. Exercisers select a speed at which they desire to exercise; maximal effort is required when the exerciser's rate of movement matches the selected speed. The speed governing feature allows for smooth and controlled muscle contraction. Injuries on isokinetic machines are very rare because of this controlled speed of movement. The fundamental basis of isokinetic training is appealing, but several disadvantages prevent its widespread use in public fitness facilities (Foran, 1985). On some types of isokinetic machines there is no resistance during the eccentric phase of the repetition. Foran (1985) indicated that optimal strength gains occur when muscle is forced to exert resistance during the eccentric as well as the concentric phase of contraction. Isokinetic machines govern the speed of movement rather than the amount of resistance, which allows the individual to "cheat" in the sense that muscular exertion can range from minimal to maximal. A great deal of time is required to change and adjust these machines for each different body part. Isokinetic machines inhibit the body's ability to develop balance due to the fixed path of movement. Only prime

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mover muscles are stressed, with no assistance from agonist muscle groups. Due to the cost and complexity of these machines, they are usually found in rehabilitation facilities (Shankman, 1984).

Isotonic strength training involves concentric as well as eccentric muscle contraction. Exercises are typically performed with variable resistance machines or free weights. Variable resistance training provides a resistance that fluctuates so that it matches a joint's ability to produce force throughout the range of motion (Weltman & Stamford, 1982). Equipment, via the use of a cam or lever system, provides greater resistance at the joint angles at which the individual is stronger and a lesser resistance at the weaker angles, therefore accommodating the lifter's strength. Variable resistance machines incorporate both concentric and eccentric muscle contraction. It is easy to evaluate progress because the lifter can increase the resistance, sets, or number of repetitions. Changing the weight is accomplished by simply moving a pin up and down the stack of weights. The machines are very safe to use because the weights move in a fixed path and for many exercises, a spotter is not necessary. As with other forms of strength training, variable resistance has its disadvantages. The resistance changes that occur throughout the range of motion are based upon averages of force-angle curves (Foran, 1985). Human beings vary in terms of limb length, therefore limiting the application of these force-angle curves in the design of variable resistance equipment. Exercises occur in a fixed path, which prevents assisting movement from agonist muscles and hinders the development of muscular coordination (balance). Finally, the machines themselves are very expensive and require a great deal of maintenance.

Free weight training involves muscular contraction against a constant resistance. Training with free weights requires the activation and coordination of antagonist in addition to prime mover muscle groups (Foran, 1985). There is not a fixed

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path of movement with free weights, which allows the exerciser to develop a sense of balance. As noted earlier, this form of strength training employs both concentric and eccentric contraction; research indicates that this combination results in more significant strength gains when compared to training either separately (Foran, 1985). Strength gains are easy to monitor, as evidenced by increases in weightloads, sets, or repetitions. Changing weights is very quick and easy to do. Free weight equipment is relatively inexpensive and requires minimum maintenance. Despite advantages, the use of free weights does have inherent limitations. When exercising against constant resistance, the muscle is only as strong as its weakest point throughout the range of motion. This deficiency is compounded when the individual employs improper technique to accomplish the desired movements. More importantly, free weight use is accompanied by an increased chance of injury. Some form of injury is almost certain to occur if proper technique and safety procedures are not observed.

Technique Implications

Proper Technique

Mastery of the proper techniques for strength training exercises is an absolute prerequisite for improving muscular strength and avoiding injury. For maximal benefit, all repetitions should be performed slowly throughout the full range of motion (Stamford, 1984). Controlling the weight, as opposed to swinging or dropping it, places maximal stress on the exercising muscle groups. Improper technique allows for assisting movement from agonist muscle groups, therefore reducing the amount of stress placed on prime mover muscle groups. Performing strength exercises against too heavy of a resistance results in poor technique. During certain phases of concentric contraction, momentum, created by body motion, assists in movement of the

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resistance rather than by the muscles exclusively. Pausing at full extension and full flexion, along with slow controlled movement reduces the risk of sustaining an injury.

Training Related Injuries

Strength training programs, in an attempt to reduce the chance of injury during athletic participation, have on numerous occasions been the cause of an injury. The presence of this dilemma might cause some individuals to question the legitimacy of strength programs as a means of injury prevention. Studies indicate that strength training injuries are the result of one or more factors. These causal factors have been identified as a lack of professional (qualified) supervision, improper technique, and overuse (Shankman, 1984). Furthermore, it has been suggested that the type of injury sustained may be related to the type of training program that is used.

Isometric training, as noted earlier, involves muscular contraction with no change in the joint angle. Injuries are nearly nonexistent in well-designed and supervised isometric programs. Injuries that occur are in the form of muscular strains that are a result of a sudden and maximal contraction (Shankman, 1984). The prevention of these strains is contingent upon performing slow and controlled contractions in which tension is gradually increased, rather than occurring instantaneously. Despite the relative safety of isometric strength training, it is very ineffective in terms of optimal strength development.

It was stated earlier in this section that isokinetic strength training is performed on machines that combine accommodating resistance with a speed governing capability. Shankman (1984) cited improper technique as the cause of muscle injury on isokinetic machines. Explosive muscle contraction against slow speed, high resistance settings places a tremendous amount of stress on the exercising joint and connective tissue. The key to injury prevention on isokinetic machines is the same as

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that of any other training protocol, slow and controlled movement throughout the entire range of motion.

Isotonic (variable resistance and free weight) modalities are the most popular, and unfortunately, the most dangerous forms of strength training. The number and severity of injuries sustained on isotonic equipment by far outnumbers other training protocols. Isotonic injuries are a result of uncontrolled eccentric movement and explosive concentric contraction against too heavy of a resistance. In addition to strain, joints and surrounding structures are subjected to excessive torque, compression and shear forces (Shankman, 1984). The bottom line to injury prevention in strength training programs is proper technique. Qualified supervision, a gradual increase in weightload, and complete adherence to proper lifting technique will facilitate maximal strength gains.

Biceps Curl Technique

A great deal of literature has been published on the mechanics of the biceps curl technique, but nothing with regard to technique deviation as a result of increased weightloads. A review of literature revealed that a consensus existed concerning proper biceps curl technique (Hesson, 1985; Kostik & Knortz, 1980; McKeon, 1990; Rasch, 1990; Riley, 1983; Schwarzenegger, 1985; Wescott, 1982). For this reason, only one author's view was included to describe the proper technique of the biceps curl. The literature suggested that the barbell curl is one of the most popular exercises for the biceps muscle group. Two points of emphasis were prevalent in the literature. Wescott (1982) stressed that the key to obtaining desired results from the exercise was contingent upon the ability of the upper arms to remain stationary while the lower arms went through the full range of motion. He emphasized that movement must occur at the elbow joint, with a minimum of assisting movement at the shoulder joint. Wescott

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stated that when the upper arms were kept stationary and the elbows were pressed into the sides, the ensuing curling movement was a result of biceps activation. Of equal importance in the literature was trunk movement. Wescott stressed the importance of maintaining an erect trunk during the execution of the barbell curl. Trunk movement, whether forward or backward, reduced the role of the biceps as the prime mover muscle group and allowed other body segments to assist in the movement. Additionally, trunk movement resulted in an increase in momentum as the weight was curled upward and further reduced the role of the biceps in the exercise. An interpretation, based upon a review of multiple sources, resulted in the following depiction of a proper biceps curl. Movement started with the arms in a fully extended position. As the weight was curled upward to a position of full flexion, the position of the elbows remained unchanged (tucked into the sides) and the trunk remained erect, having placed all of the stress on the biceps muscle groups. The repetition was completed by slowly lowering the weight back to the starting position.

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CHAPTER III

DESIGN AND METHODOLOGY

The purpose of this study was to identify the deviations in proper biceps curl technique that occur as a result of the varying intensities of an individual's 1 RM. The chapter is organized into four content areas: (1) subject selection, (2) instrumentation, (3) experimental design, (4) data analysis and (5) statistical procedures.

Subject Selection

Subjects that participated in this study were ten male college students between the ages of 19 and 23 years. During the data collection phase of the investigation, the participants were enrolled in general physical education weight training courses. Prior to testing, announcements regarding the study were made to various sections of weight training classes and interested students contacted the researcher. The subjects that comprised the data-producing sample were then randomly selected from the general population of respondents.

The study was approved by the Human Subjects Review Board at Western Michigan University. The approval form is presented in Appendix A. Prior to data collection, the subjects were questioned with respect to their health status. Each of the subjects indicated that there were no health risks present that would prevent them from performing the testing procedure. A consent form was signed and dated by each subject. Appendix B contains a copy of the consent form.

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Instrumentation

Several pieces of biomechanical instrumentation were necessary to ensure accurate data collection and valid analysis. The instruments that were used in this study have been used repeatedly in other studies that involved movement analysis.

Photo-Sonic 1PL Camera

The filming was done with a Photo-Sonic 1PL camera. This high speed 16 mm camera was specifically designed to film rapid movement events. The frame rate for this camera ranges between 10 and 500 pictures per second. The developed film allows the viewer to perform a frame-by-frame analysis of the movement. This instrument was calibrated prior to filming, to ensure a reliable and accurate method of data collection. The calibration procedure of the camera centered around correct film speed. The accuracy of the selected film speed was contingent upon two factors. Prior to filming the camera's power pack was fully charged. In addition, light emitting diodes (LED's) were exposed on the film during filming. The LED's were used as a method of calibrating film speed.

Vanguard Motion Analyzer

The film analysis phase of the project required the use of a Vanguard motion analyzer, serial number 23752, equipped with projection head M-16C. The motion analyzer is designed so that a frame-by-frame analysis of the movement pattern can be made.

Peripheral Instrumentation

Additional instrumentation was necessary for film analysis. The film digitizing process

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was completed with a Numonics Electronic Graphics calculator that was interfaced with a computer. Data analysis was done with the use of Peak 3-D software, version 4.5 (Peak Performance Technology IES, Inc., 1989). Before the digitizing process began, a reference measure filmed prior to the treatment procedures, was identified and true vertical and horizontal measures were established so that the input data would be accurate. To provide adequate lighting, six banks of Colortron lights were used during filming. The use of these lights eliminated shadows, which made it easier to detect joint movement during film analysis.

Experimental Procedures

All data were collected in one afternoon. To test the research problem that varying intensities of the 1 RM elicit biomechanical movements that are incorrect with regard to proper biceps curl technique, a repeated measures design was implemented. A randomized order of testing was incorporated in this design, which allowed for more valid data production. The subjects performed each of the five conditions in a random order. The selected conditions for this study were 60% (Condition 1), 70% (Condition 2), 80% (Condition 3), 90% (Condition 4), and 100% (Condition 5) of each individual's 1 RM.

The subjects were filmed with the Photo-Sonic 1PL camera set at a speed of 100 fps (feet per second) from a distance of approximately 25 feet. For this particular study, only the concentric phase of each condition was filmed. The subjects were all filmed with the right side of their body facing the camera. The filming of the subjects was done against a black background to help facilitate the viewing of body movement during film analysis. Six banks of Colortron lights were aimed directly at the subjects to provide adequate lighting for filming. Prior to filming, a reference measure, a yardstick, was filmed for analysis purposes. Every subject performed a complete

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repetition at each of five pre-determined intensities. For each of the repetitions, subjects initiated movement upon reception of a verbal command from the camera operator. At the completion of each repetition the necessary weightload adjustments were made and the procedure was repeated.

The film analysis took place in Western Michigan University's biomechanics laboratory. The film was loaded into the Vanguard motion analyzer and projected down on a white table. Before film digitizing began, the reference measure, taken prior to the testing procedure during filming, was located and a scale factor for frame analysis was set. Each condition for every subject was viewed in its entirety and key frames for the digitizing process were selected. Each selected frame was projected on the table and 13 anatomical landmarks were plotted on an X-Y coordinate graph. The landmarks representing joint centers that were plotted consisted of the following: (1) distal point of the right hand, (2) right ulna styloid, (3) right center of elbow, (4) right coracoid process, (5) distal point of the right foot, (6) right medial malleolus, (7) right center of knee, (8) right greater trochanter, (9) top of sternal notch, (10) pubic bone, (11) tragus of the ear, (12) top of head, and (13) center of object.

Once each landmark was located, the X-Y coordinate was entered with the Numonics Electronic Graphics calculator and the data were entered into the interfaced computer. This process was repeated for all of the selected frames. At the conclusion of each treatment, the subject's weight, the object's weight and the time between frames was entered and the data were stored.

Data Analysis

Data analysis was completed by using a repeated measures design. The dependent variable was defined as the range of motion in the sagittal plane during the various phases of the biceps curl. The independent variables were defined as the

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different weightload intensities of an individual's 1 RM: 60%, 70%, 80%, 90%, and 100%. Each subject performed the conditions in a random order. For example, Subject 1 may have performed Condition 4 first, while Subject 2 may have performed Condition 1 first. Presenting the conditions in a random order provided internal validity (the degree to which the independent variable caused change in the dependent variable) for the study. More specifically, error such as testing effect, was minimized in the study. Appendix C contains a listing of the order in which each subject performed the conditions.

Data comparisons among subjects were made to determine the magnitude of differences with respect to body mechanics that occurred for each condition. Of particular interest were the angular changes that occurred at the shoulder, trunk, body, and knee joints. In order to calculate the angle for a specific joint, it was necessary to use the X-Y coordinate values of three anatomical locations. P1 and P2 were the segmental endpoints of a particular angle and V (vertex) was the point of intersection of the two sides of the angle. Appendix D contains a listing of the anatomical landmarks that were used for the calculation of each joint angle. The shoulder angle was calculated with the X-Y coordinate values of the right elbow (P1), the right shoulder (V) and the horizontal (P2). The body angle was calculated with the X-Y coordinate values of the right shoulder (P1), the right ankle (V), and the vertical (P2). The X-Y coordinate values of the right wrist (P1), the right elbow (V), and the right shoulder (P2), were used to calculate the elbow angle. The trunk angle was calculated with the X-Y coordinate values of the right shoulder (P1), the right hip (V), and the vertical (P2). The X-Y coordinate values of the right hip (P1), the right knee (V), and the right ankle (P2) were used to calculate the knee angle.

The angular position of the elbow during the execution of the biceps curl was used as a reference measure to select key frames for analysis and comparison. Five

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positions were selected to analyze biomechanical deviations that occurred in response to the various conditions. The first position selected was the actual initiation of the biceps curl. The second position selected for analysis was at 135° elbow flexion. An elbow angle of 90° was selected as the third position for data analysis. The fourth position was at 45° elbow flexion. The final position selected for analysis was at the completion of all body movement. The initiation and completion positions were necessary in order to determine the magnitude of change that occurred at selected joints during the execution of the biceps curl. The position of 90° elbow flexion was selected because it has been referred to as the "sticking point" during the repetition. In other words, it is at this position that the body deviates in order to facilitate movement. The positions of 135° and 45° elbow flexion were selected because they hypothetically fall half way between 90° and their respective initiation and completion positions.

Statistical Procedures

Several statistical procedures were used to describe the data that was generated in the study. The mean of the group for each of the selected angles was used to compare the differences among the five conditions. Descriptive statistics were used to describe patterns of motion for each joint angle. F-ratios were used to determine variability between the conditions. The independent variables were: (a) the five conditions, 60%, 70%, 80%, 90% and 100% of the subject's 1 RM; and (b) the randomized treatment order of the five conditions. The dependent variables that were measured were changes in: (a) shoulder angle, (b) trunk angle, (c) body angle, and (d) knee angle.

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CHAPTER IV

RESULTS AND DISCUSSION

This chapter summarizes the results obtained from the present study and discusses the implications of the results in identifying an optimal training intensity for the biceps curl. The purpose of this study was to identify the biomechanical deviations that occurred in response to increased weightload intensities, thereby making it possible to identify an optimal training intensity for the biceps curl. It was the intention of this paper to help alleviate some of the confusion surrounding training intensities in the field of strength training by providing biomechanical evidence that supports the contention that a specific training intensity will facilitate proper strength training technique and yield maximal gains in terms of strength improvement.

The following raw data were used to analyze the biomechanical deviations that resulted from increased weightload intensities: (a) shoulder angle, (b) trunk angle, (c) body angle, and (d) knee angle. The generated data were analyzed on an individual and group basis. For every subject, comparisons were made among conditions to identify significant changes in the selected anatomical parameters. Comparisons among conditions with respect to the group of subjects as a whole were made by calculating the mean, the variance, and the F-ratio from the raw data. The descriptive data of each subject for all five conditions are presented in Appendix E.

Results

For descriptive analysis purposes the motion of the arm curl was divided into four phases. Phase I began with the initiation of the curl and ended when the elbow

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reached 135°. Phase II began when the elbow angle was 135° and ended when the elbow reached 90°. Phase III began when the elbow angle was 90° and ended when the elbow reached 45°. Phase IV began when the elbow angle was 45° and ended when the elbow reached its greatest degree of flexion. These phases were used to discuss and describe the mechanics of the biceps curl throughout this chapter.

Homogeneity of Variance

F-ratios were calculated to determine if homogeneity, with respect to variance, existed between the five conditions. In this particular study only variances for the extreme conditions were compared. More specifically, the 60% RM condition was compared to the 70% RM, 80% RM, 90% RM, and 100% RM conditions. Only the comparisons in which there was an absence of homogeneity were addressed. The homogeneity of variance comparisons (see Table 1) indicated the following:

1. An absence of homogeneity, F=3.34, existed between Condition 1 and Condition 2 in the shoulder angle at 135° elbow flexion ($F(9,9)$=3.16, p<.05).

2. An absence of homogeneity, F=3.65, existed between Condition 1 and Condition 2 in the trunk angle at 135° elbow flexion ($F(9,9)$=3.16, p<.05).

3. An absence of homogeneity, F=4.84, existed between Condition 1 and Condition 2 in the body angle at 135° elbow flexion ($F(9,9)$=3.16, p<.05).

4. An absence of homogeneity, F=4.0, existed between Condition 1 and Condition 3 in the body angle at 135° elbow flexion ($F(9,9)$=3.16, p<.05).

5. An absence of homogeneity, F=3.72, existed between Condition 1 and Condition 4 in the trunk angle at the initiation of movement ($F(9,9)$=3.16, p<.05).

6. An absence of homogeneity, F=6.6, existed between Condition 1 and Condition 4 in the trunk angle at 90° elbow flexion ($F(9,9)$=3.16, p<.05).

7. An absence of homogeneity, F=4.12, existed between Condition 1 and

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Condition 4 in the body angle at 135° elbow flexion ($F(9,9)=3.16$, $p<.05$).

8. An absence of homogeneity, $F=6.66$, existed between Condition 1 and Condition 4 in the body angle at 90° elbow flexion ($F(9,9)=3.16$, $p<.05$).

9. An absence of homogeneity, $F=4.08$, existed between Condition 1 and Condition 5 in the trunk angle at 135° elbow flexion($F(9,9)=3.16$, $p<.05$).

10. An absence of homogeneity, $F=7.95$, existed between Condition 1 and Condition 5 in the trunk angle at 90° elbow flexion ($F(9,9)=3.16$, $p<.05$).

11. An absence of homogeneity, $F=8.8$, existed between Condition 1 and Condition 5 in the trunk angle at 45° elbow flexion ($F(9,9)=3.16$, $p<.05$).

12. An absence of homogeneity, $F=5.13$, existed between Condition 1 and Condition 5 in the body angle at 45° flexion ($F(9,9)=3.16$, $p<.05$).

13. An absence of homogeneity, $F=4.19$, existed between Condition 1 and Condition 5 in the body angle at the completion of movement ($F(9,9)=3.16$, $p<.05$).

14. An absence of homogeneity, $F=4.43$, existed between Condition 1 and Condition 2 in the knee angle at 45° elbow flexion ($F(9,9)=3.16$, $p<.05$).

Shoulder Angle

Three landmarks were used to calculate the shoulder angle, two of these landmarks were endpoints and one was the vertex. The right elbow was the first endpoint (P1), the right shoulder was the vertex (V), and the horizontal was the second endpoint (P2). For this angle, 90° represented the anatomical position. Angular values greater than 90° indicated extension, while values less than 90° indicated flexion. Based upon the generated shoulder angle data the subjects were grouped with respect to similar movement patterns that were maintained across all of the conditions.

For the shoulder angle three distinct movement patterns were identified. Subjects 1, 7, and 9 comprised one of these three groups. In Phase I and Phase II the

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Table 1

Summary Table for Homogeneity of Variance

Joint	Initiation	135°	90°	45°	Completion
		Condition 1 VS. Condition 2			
Shoulder	1.14	3.34*	1.48	.67	.79
Trunk	1.25	3.65*	1.85	.53	.89
Body	2.52	4.84*	2.31	.02	2.10
Knee	1.74	1.70	1.58	1.22	.88
		Condition 1 VS. Condition 3			
Shoulder	1.33	2.04	1.96	.62	.47
Trunk	1.06	2.99	1.89	2.21	1.32
Body	2.59	4.00*	2.94	.03	2.85
Knee	1.43	.92	1.09	1.07	1.15
		Condition 1 VS. Condition 4			
Shoulder	1.71	3.05	2.97	.93	.69
Trunk	3.72*	3.01	6.60*	2.75	1.52
Body	2.37	4.12*	6.66*	.05	2.94
Knee	2.72	1.57	1.29	1.09	1.09
		Condition 1 VS. Condition 5			
Shoulder	2.12	2.29	2.34	.30	.43
Trunk	1.53	4.08*	7.95*	8.80*	1.26
Body	1.60	1.28	5.13	.14	4.19*
Knee	1.95	1.98	1.36	4.43*	1.73

* $F(9,9) = 3.18$, $p < .05$

shoulder extended, as indicated by the progressive increase in the shoulder angle values. In Phase III and Phase IV the direction of shoulder movement changed and the shoulder flexed, evidenced by the decrease in shoulder angle values. In Phase III there was a dramatic decrease in the shoulder angle, while the decrease in Phase IV

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was less extensive. An example of the type of movement pattern representative of this first group is illustrated in Figure 1.

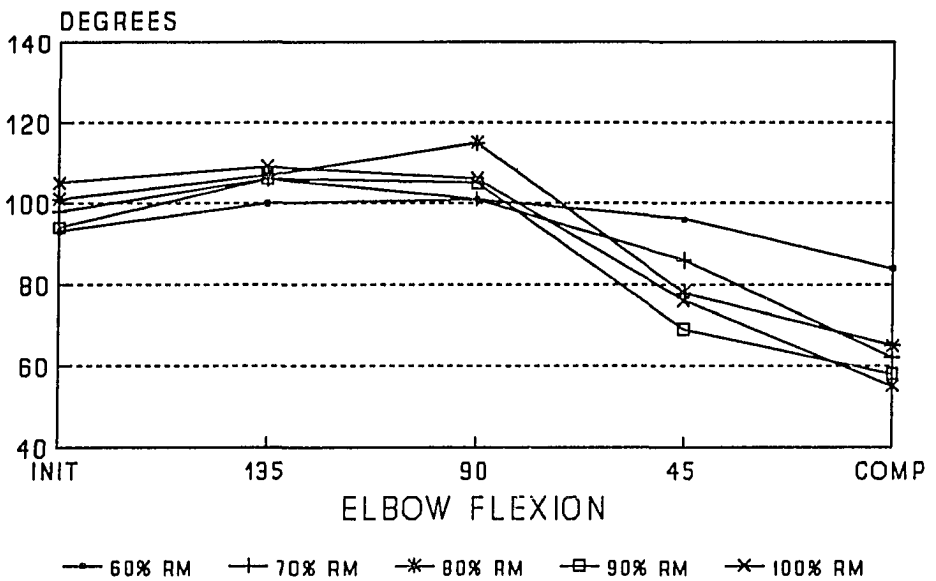


Figure 1. Subject 7 Shoulder Angle Movement Pattern.

For descriptive purposes, the 80% RM condition of Subject 7 will be described in detail. In Phase I, the shoulder extended 6.8°, from 100.6°, to 107.4°. In Phase II the shoulder extended an additional 7.2° to an angle of 114.6°. In Phase III the shoulder flexed dramatically, 36.4°, to a position of 78.2°. In Phase IV the shoulder continued to flex an additional 13.7° to a final angle of 64.5°.

Subjects 2 and 8 displayed a second type of movement pattern with respect to the shoulder joint during the execution of the biceps curl. In Phase I the position of the shoulder remained relatively unchanged, as indicated by the negligible change in the shoulder angle. In Phase II, Phase III, and Phase IV, the shoulder had been very active, having flexed, as indicated by the progressive decrease in the shoulder angle values. The changes that occurred in the shoulder angle in Phases II and III were

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comparable in magnitude. The shoulder angle continued to decrease in Phase IV, however, the changes were smaller in magnitude in comparison to the decreases observed in the preceding phases of the biceps curl. Figure 2 illustrates the type of movement pattern observed for this group.

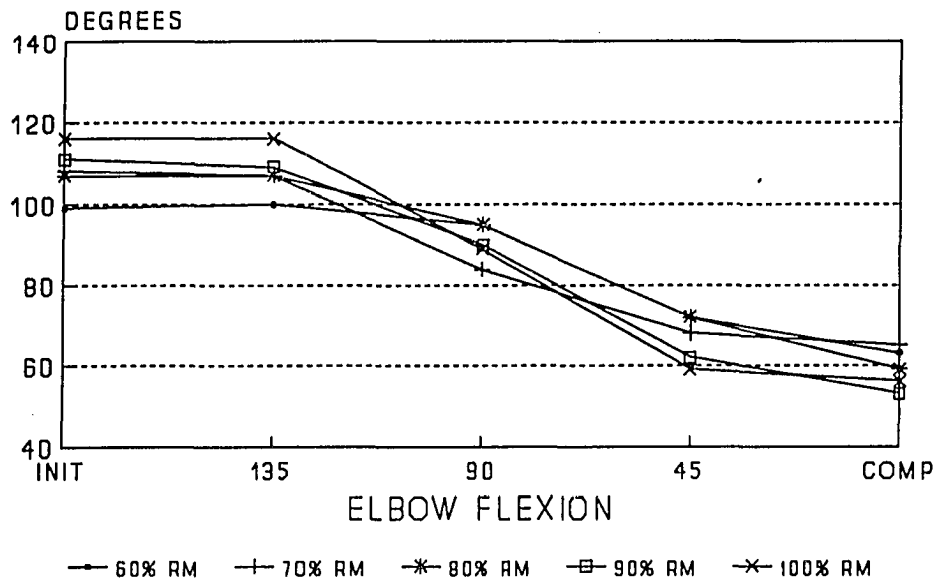


Figure 2. Subject 2 Shoulder Angle Movement Pattern.

For descriptive purposes, the 90% RM condition performed by Subject 2 will be described in detail. In Phase I the shoulder flexed a negligible 2.4°, from 111.3° to an angle 108.9°. In Phase II the shoulder flexed extensively, 19.3°, to a position of 89.6°. A comparable flexion of 27.6° to an angle of 62.° occurred in Phase III. A considerably smaller flexion, 8.6°, to a final angle of 53.4° resulted in Phase IV.

Subjects 3, 4, 5, 6, and 10 comprised the third movement pattern observed at the shoulder joint. In Phase I the shoulder extended, similar to the first group, as indicated by the increase in shoulder angle values. In Phase II, Phase III, and Phase IV the shoulder flexed, as indicated by the decrease in shoulder angle values, however,

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the extent to which the shoulder angle decreased varied in magnitude during the various phases of the biceps curl. In Phase II Subjects 6 and 10 exhibited a more overt change in the shoulder angle, while Subjects 3, 4, and 5 demonstrated a less severe change in the shoulder angle. All five subjects revealed smaller changes in the shoulder angle in Phase III. The changes that occurred in the shoulder angle in Phase IV were smaller in magnitude when compared to the previous phases. Figure 3 illustrates the type of movement pattern observed for this group.

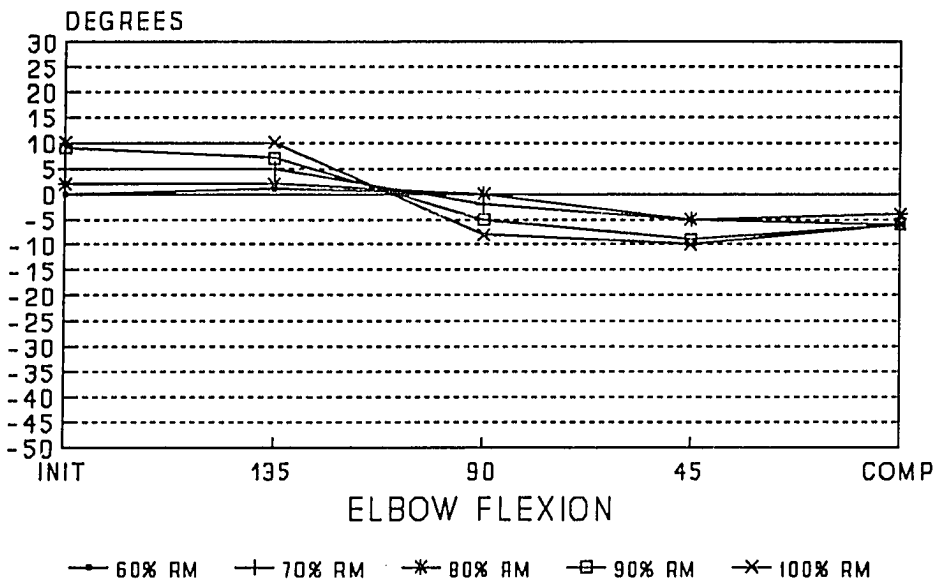


Figure 3. Subject 3 Shoulder Angle Movement Pattern.

For descriptive purposes, the 90% RM condition performed by Subject 3 will be described in detail. In Phase I the shoulder extended 12.4°, from 89.8°, to an angle of 102.2°. A flexion of 10.8° to a position of 91.4°, occurred in Phase II. A more overt flexion, 22.9°, to an angle of 68.5° resulted in Phase III, and in Phase IV a substantially smaller flexion of 7.7° to a final angle of 60.8° occurred.

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Trunk Angle

Three landmarks were used to calculate the trunk angle, two of these landmarks were endpoints and one was the vertex. The right shoulder was the first endpoint (P1), the right hip was the vertex (V), and a vertical line represented the second endpoint (P2). An angle of 0° represented the anatomical position. Angular values greater than 0° represented flexion, while values less than 0° indicated extension. Based upon the observed movement at the trunk angle the 10 subjects were placed into four groups.

Subjects 2 and 8 comprised one of these four groups. In Phase I very little movement occurred at the trunk, as indicated by the negligible changes in the trunk angle. In Phase II the changes that occurred in the trunk angle were present only in the 90% RM and 100% RM conditions, in which the trunk extended. The 60% RM, 70% RM, and 80% RM conditions resulted in little or no change in the position of the trunk. In Phase III the trunk extended in all five conditions, as indicated by the decrease in trunk angle values. In Phase IV the direction of trunk movement changed, having flexed slightly, evidenced by the increase in trunk angle values. Figure 4 illustrates the type of trunk movement observed for this particular group.

For descriptive purposes, the 90% RM condition performed by Subject 2 will be described in detail. In Phase I the trunk extended only 2.0°, from 8.9° to an angle of 6.9°. However, in Phase II the trunk extended extensively, 11.7°, to an angle of -4.8°. An additional extension of 4.2° to an angle of -9.0° occurred in Phase III. In Phase IV the direction of movement changed and the trunk flexed 3.0° to a final angle of -6.0°.

A second classification with respect to trunk movement consisted of Subjects 1 and 9. With the exception of the 100% RM condition very little movement and subsequent change occurred in the trunk angle throughout the execution of the

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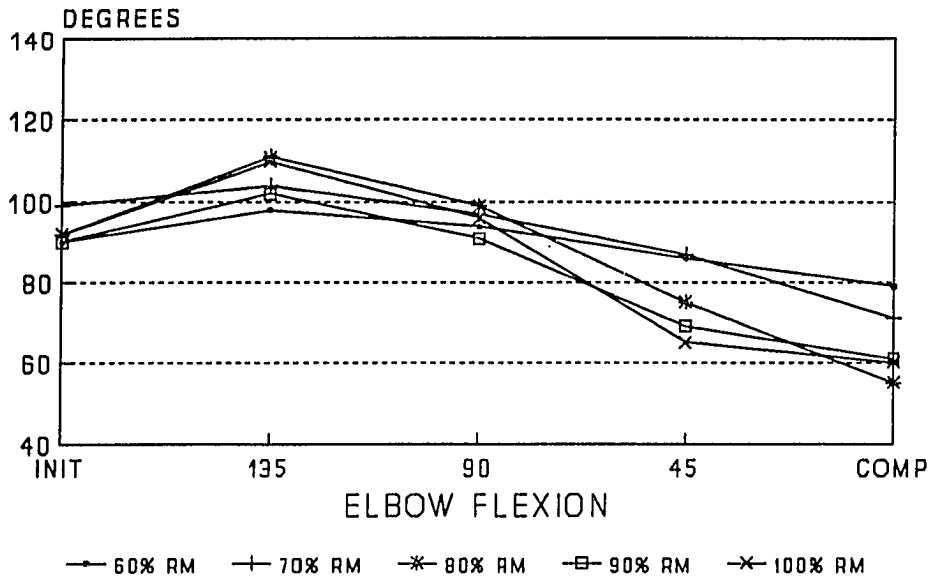


Figure 4. Subject 2 Trunk Angle Movement Pattern.

biceps curl. In Phase I for the 60% RM, 70% RM, 80% RM, and 90% RM conditions the trunk flexed only slightly, as indicated by the minimal increases in trunk angle, while in the 100% RM condition, flexion ofthetrunk was more pronounced. In Phase II the direction of trunk movement changed and the trunk extended to approximately the same position as when movement was first initiated. In Phase III the trunk continued to extend, with movement more overt in the 100% RM condition. In Phase IV the motion of extension persisted, comparable in magnitude to the movement observed in the two previous phases. Figure 5 illustrates the type of trunk movement observed for this group.

For descriptive purposes, the 90% RM condition performed by Subject 9 will be described in detail. In Phase I the trunk flexed only 1.7°, from 7.7° to an angle of 9.4°. In Phase II the trunk extended 1.7° to the initial starting position of 7.7°. In Phase III the trunk continued to extend 5.5° to an angle of 2.2°. In Phase IV the trunk extended an

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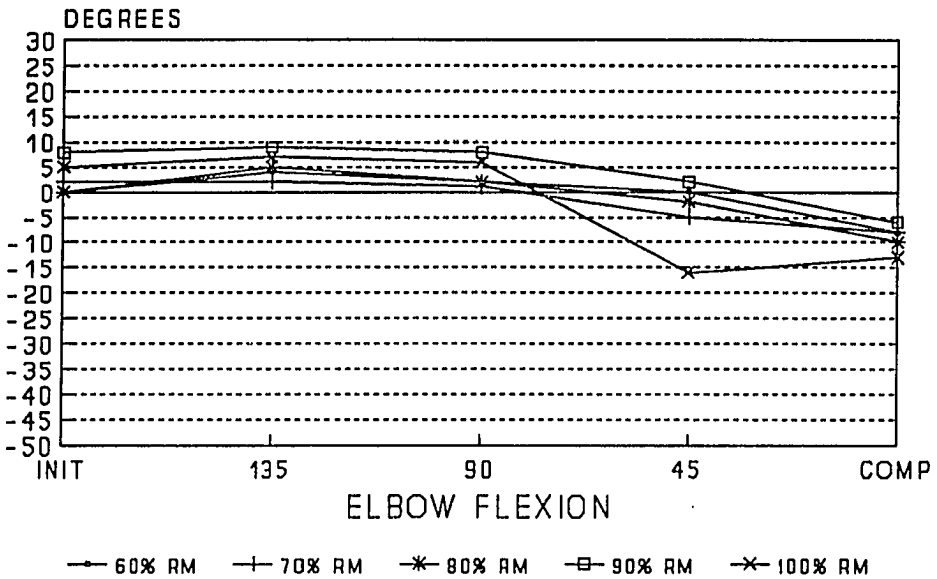


Figure 5. Subject 9 Trunk Angle Movement Patterns.

additional 8.6° to a final angle of -6.4°.

Subjects 3, 4, and 7 represented the third type of movement pattern observed in trunk motion. The magnitude of movement among the five conditions was more diverse upon examination. In Phase I trunk movement was negligible in the 60% RM, 70% RM, and 80% RM conditions, while flexion was more pronounced in the 90% RM and 100% RM conditions. In Phase II, in the 90% RM and 100% RM conditions the trunk extended substantially, while in the 60% RM, 70% RM, and 80% RM conditions trunk movement was negligible. The type of trunk movement observed in Phase III was comparable to that of the previous phase, with extension quite pronounced in the two heaviest conditions, and only minimal extension observed in the other three conditions. In Phase IV trunk movement changed direction in the 90% RM and 100% RM conditions with the trunk having flexed, while in the other three conditions the slight, yet progressive extension in the trunk persisted. Figure 6 illustrates the type of movement

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observed for this group.

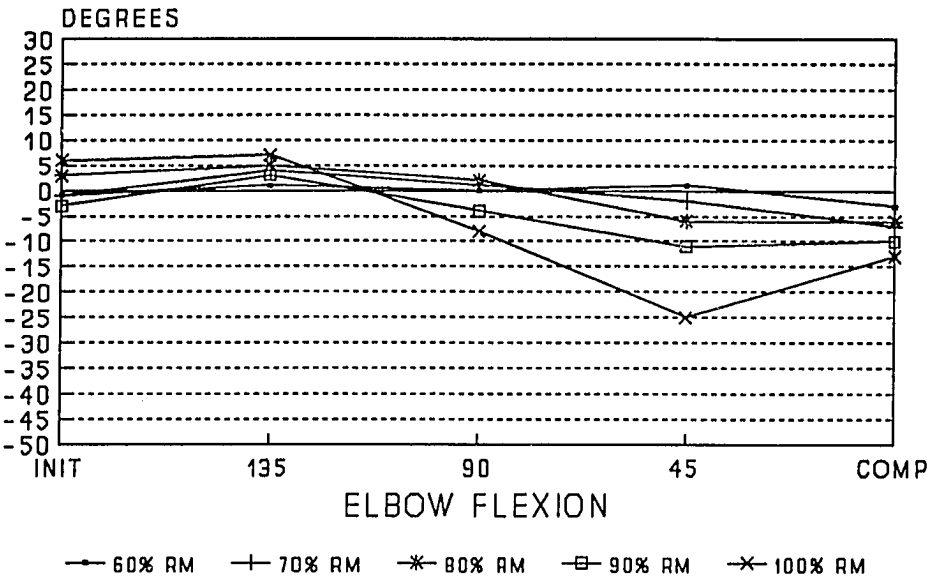


Figure 6. Subject 7 Trunk Angle Movement Pattern.

For descriptive purposes, the 70% RM and 100% RM conditions of Subject 7 will be described in detail. In Phase I of the 70% RM condition the trunk flexed only 4.6°, from -1.0° to an angle of 3.6°. In Phase II the trunk extended 2.7° to a position of 0.9°. In Phase III the trunk continued to extend slightly, 2.7°, to an angle of -1.8°. In Phase IV the trunk angle extended an additional 5.0° to a final position of -6.8°. In Phase I of the 100% RM condition the trunk flexed 1.6° from 5.5° to an angle of 7.1°. In Phase II the trunk extended dramatically, 15.4°, to an angle of -8.3°. In Phase III the trunk continued to extend an additional 19.4° to a position of -24.7°. In Phase IV the direction of trunk movement changed and the trunk flexed 11.4° to a final angle of -13.3°.

Subjects 5, 6, and 10 comprised the fourth group with respect to trunk movement. In Phase I the trunk flexed, as indicated by the increase in trunk angle

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values. During this phase of movement trunk flexion was more pronounced in the 90% RM and 100% RM conditions. In Phase II trunk movement changed direction and the trunk extended as evidenced by the decrease in the trunk angle values. Again, movement was more pronounced in the 90% RM and 100% RM conditions. In Phase III the trunk continued to extend, however, the extension was less in magnitude in comparison to the previous phrase. In Phase IV trunk movement again changed direction and the trunk flexed. Figure 7 illustrates the pattern of movement for this group.

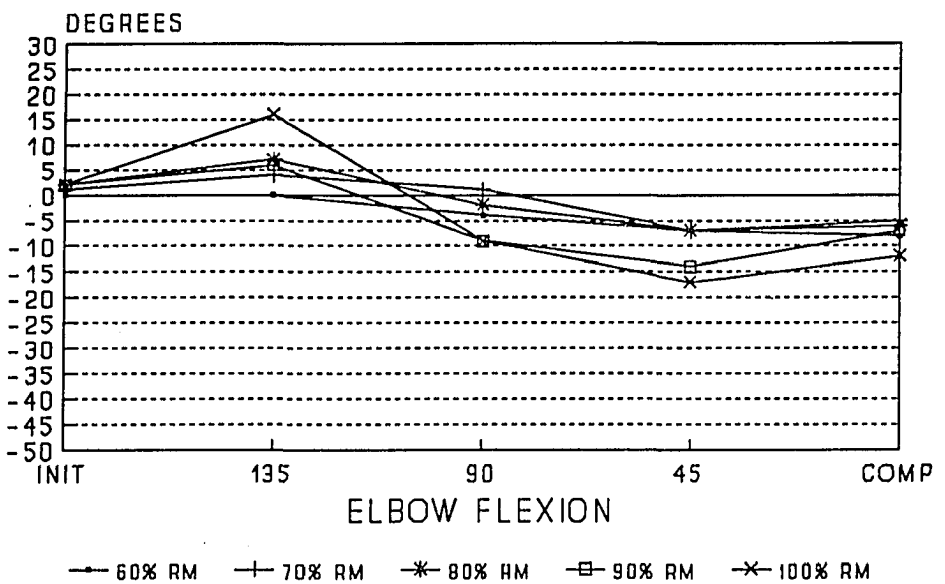


Figure 7. Subject 5 Trunk Angle Movement Pattern.

For descriptive purposes, the 90% RM condition of Subject 5 will be described in detail. In Phase I the trunk flexed 3.7°, from 1.9° to an angle of 5.6°. In Phase II the trunk extended 14.4° to a position of -8.8°. In Phase III the trunk continued to extend 4.8° to an angle of -13.6°. In Phase IV the direction of movement changed and the trunk flexed 6.4° to a final angle of -7.2°.

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Body Angle

Three landmarks were used to calculate the body angle, two of the landmarks were endpoints and the other was the vertex. The right shoulder was the first endpoint (P1), the right ankle was the vertex (V), and a vertical line represented the second endpoint (P2). An angle of 0° represented the anatomical position. An angular value greater than 0° indicated flexion, while a value less than 0° indicated extension. With respect to the body angle throughout the execution of the biceps curl, movement was quite diverse among the 10 subjects. The classification process resulted in four distinct types of movement patterns for the body angle.

Subjects 3, 4, and 7 comprised one of these four groups. In Phase I the change in body angle was minimal for the 60% RM, 70% RM, and 80% RM conditions, while a more overt flexion of the body was observed in the 90% RM and 100% RM conditions. In Phase II the body extended, as indicated by the decrease in body angle values. In Phase III the type of body movement and subsequent change that was observed was, in general, comparable to that of the previous phase. In Phase IV the observed body movement was similar to the type of movement observed in the initial phase. The 60% RM and 70% RM conditions revealed little or no change in the body angle, while in the 80% RM, 90% RM, and 100% RM conditions the body flexed, as indicated by the increase in body angle values. Figure 8 illustrates the pattern of movement observed for this group.

For descriptive purposes, the 80% RM condition of Subject 7 will be described in detail. In Phase I the body flexed only 0.6°, from 2.5° to an angle of 3.1°. In Phase II the direction of body movement changed and the body extended 1.8° to a position of 1.3°. In Phase III the body continued to extend an additional 3.9° to an angle of -2.6° . In Phase IV the direction of body movement again changed, and the body

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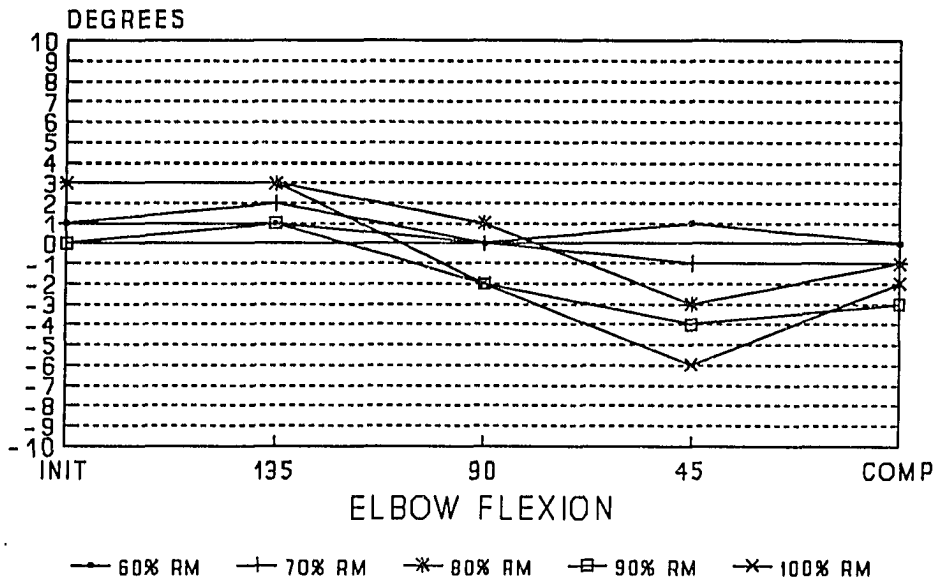


Figure 8. Subject 7 Body Angle Movement Pattern.

flexed 1.4° to a final position of -1.2°.

Subjects 5, 6, and 10 comprised the second type of movement pattern with respect to the body angle. In general, in Phase I the body flexed, as indicated by the increase in body angle. In Phase II the direction of movement changed and the body extended, evidenced by the decrease in body angle. In Phase III two types of movement patterns were observed. Subject 5 continued to extend slightly, while subjects 6 and 10 changed direction and flexed, as indicated by the increase in body angle values. In Phase IV all three subjects exhibited the same pattern of movement. The direction of movement changed for subject 5, evidenced by the increase in body angle, while subjects 6 and 10 continued to demonstrate increases in body angle values. Figure 9 illustrates the pattern of movement observed for this group.

For descriptive purposes, the 80% RM condition for Subject 6 will be described. In Phase I the body flexed 4.2°, from 1.6° to an angle of 5.8°. In Phase II the direction

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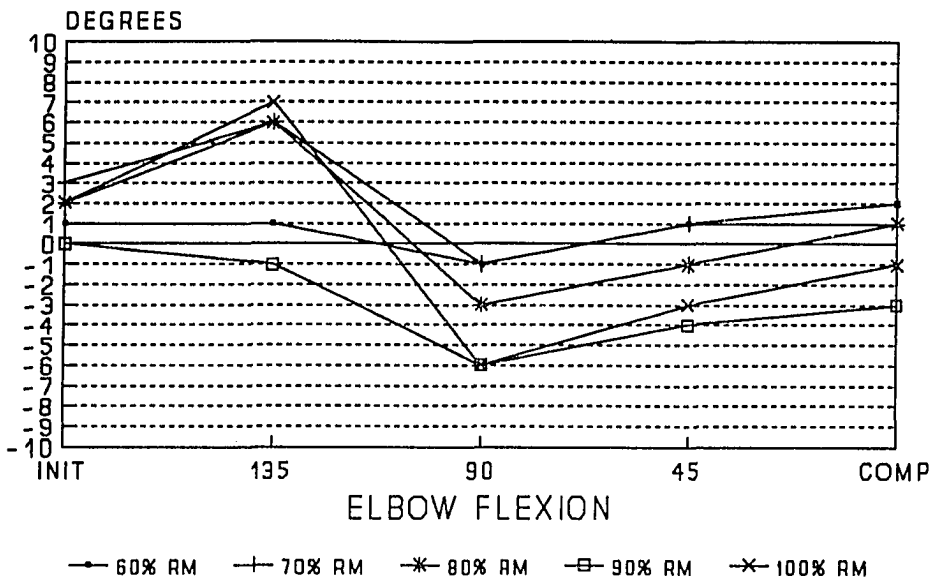


Figure 9. Subject 6 Body Angle Movement Pattern.

of movement changed, and the body extended, 8.4°, to a position of -2.6°. In Phase III the direction of body movement again changed and the body flexed 1.6° to an angle of -1.0°. In Phase IV the body flexed an additional 1.5° to a final angle of 0.5°.

Subjects 1, 2, and 8 comprised the third type of movement identified for the body angle. In general, body movement in Phase I was negligible. In Phase II the body extended, as indicated by the decrease in the body angle values. In Phase III movement was quite diverse. All three subjects demonstrated both flexion and extension. As a result, no distinguishable pattern could be identified, because while one subject may have demonstrated extension in the body angle in a specific condition another subject may have demonstrated flexion. However, in Phase IV all three of the subjects demonstrated flexion in the body angle, evidenced by the increase in body angle values. Figure 10 illustrates the movement pattern observed for this group.

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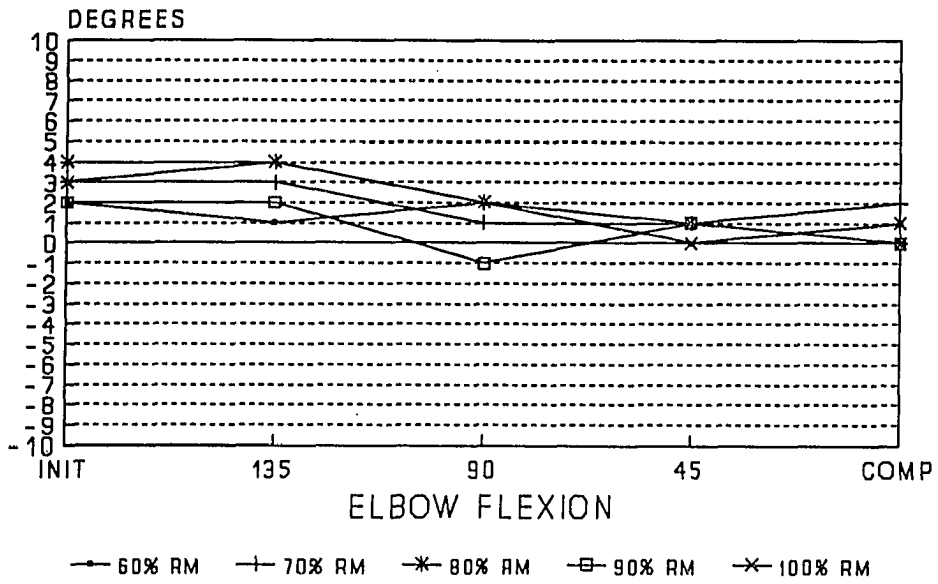


Figure 10. Body Movement Across Treatment Conditions for Subject 1.

For descriptive purposes, the 90% RM condition of Subject 1 will be described. In Phase I the body flexed only 0.4°, from 1.8° to 2.2°. In Phase II the body extended 2.9° to a position of -0.7°. In Phase III the direction of movement changed and the body flexed 1.2° to an angle of 0.5°. In Phase IV the direction of movement again changed and the body extended 0.2° to a final angle of 0.3°.

The changes observed in the body angle for Subject 9 were not similar to any of the previously discussed groups, therefore this subject was classified as the fourth type of movement pattern. In Phase I there was no observed change in the body angle. In Phase II there were no observed changes in the 60% RM and 100% RM conditions, while there was a slight extension observed in the 70% RM, 80% RM, and 90% RM conditions. In Phase III the body extended in all five conditions, with the change more overt in the 100% RM condition. In Phase IV the body continued to extend in all but the 100% RM condition, in which body movement changed direction,

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as indicated by the increase in body angle values. Figure 11 illustrates the movement pattern observed for Subject 9.

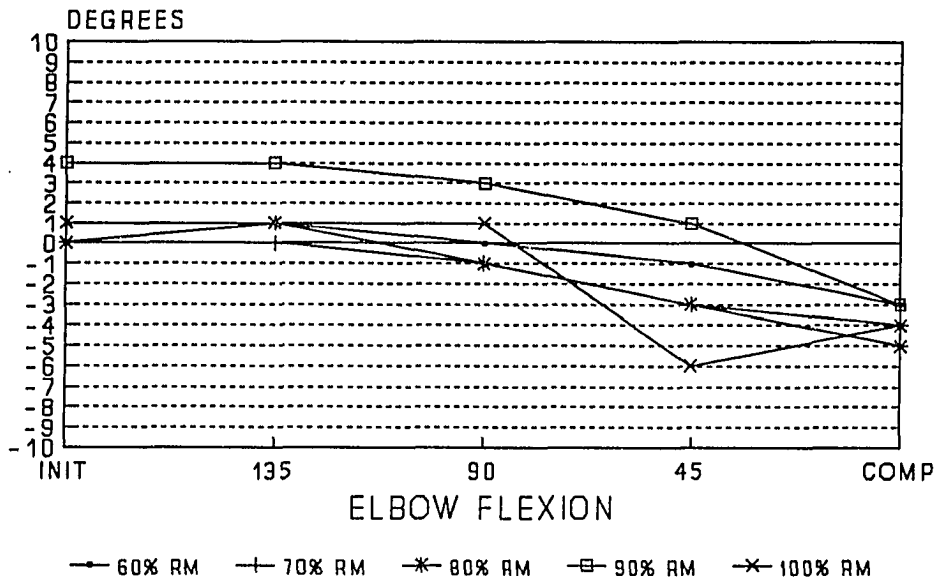


Figure 11. Subject 9 Body Angle Movement Pattern.

For descriptive purposes, the 80% RM condition will be described. In Phase I the body flexed only 0.6°, from 0.0°, to an angle of 0.6°. In Phase II the body extended 1.5° to a position of -0.9°. In Phase III the body continued to extend 1.6° to an angle of -2.5°. In Phase IV the body extended an additional 2.0° to a final angle of -4.5°.

Knee Angle

The subjects demonstrated a wide array of movement about the knee joint for the five conditions. Movement among conditions, and more specifically the phases, during the execution of movement was very diverse and unpatterned. For this reason no attempt was made to categorize the types of movement that were observed. Several observations, however, could be made with respect to the movement that

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occurred at the knee joint. In general, the resultant changes in knee angle from Phase I to Phase IV were not substantial even though change during various phases of the biceps curl was extensive. Moreover, with regard to the observed overall changes in the knee angle, there were a comparable number of increases, decreases, or no changes among the 10 subjects. Finally, in the 90% RM and 100% RM conditions knee movement was more extensive, as indicated by the more overt increases and decreases in knee angle values.

Discussion

The subjects in this study were college age males with limited experience in the area of strength training. The establishment of each subject's 1 RM revealed a range from 85 lbs to 135 lbs. The established 1 RM values comprised quite a diverse range, therefore the results of the study should be applicable to the majority of college age males. All subjects underwent identical testing procedures. The only distinguishing factor between subjects was the order in which the five conditions were performed. Each subject performed the conditions in a random order to eliminate any biases. Rather than expound upon the causes of individual differences in the selected angles, the reasons attributed to changes in mean values for each of the selected angles are discussed. The group means for the selected angles are presented in Appendix F.

Shoulder Angle

Examination of shoulder angle mean values for each of the five conditions revealed a similar movement pattern throughout the selected phases of the biceps curl. In Phase I, with the exception of the 100% RM condition, the shoulder joint extended, as indicated by the increase in the shoulder angle values. This type of movement at the shoulder joint may have been a compensatory adjustment made

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by the body in an attempt to overcome the inertia of the stationary resistance. Strength training literature suggested that this response is observed in numerous strength training exercises and that as resistance is increased the magnitude of compensatory adjustment made by the body correspondingly increases.

In Phase II the direction of shoulder movement changed and the shoulder joint flexed, as indicated by the decrease in shoulder angle means. Additionally, this type of movement was observed in all five conditions and persisted in the remaining phases of the biceps curl, with the magnitude of flexion greater in the heavier resistance conditions. This type of joint response is attributed to a combination of factors. The first influencing factor pertains to the lever-like arrangement of the elbow joint and the torque production capabilities of the elbow flexors. Torque production is contingent upon two factors, a force (muscle) and a force arm (bone). The torque production of any force is dependent on the angle of applied force to the force arm and also the length of the force arm. Manipulation of the force arm (lengthening/shortening) influences the torque production capabilities of the force. The largest force arm that can be achieved by the elbow flexors at the elbow is that produced when the force (muscle) pulls at a 90° angle to the bones. The effect of changing a muscle's angle of pull influences the ability of the muscle to produce torque. If the angle of pull is less than 90° (elbow extension) the muscle has rotary (a part of the force that is perpendicular to and acts to rotate the bony lever) and stabilizing (a part of the force that is parallel to the bone and acts toward the axis of rotation) components. If the muscle pulls at a 90° angle (elbow flexion) the muscle force is 100% rotary. When the angle of pull of the muscle is greater than 90° (continued elbow flexion) the rotary component of force diminishes and the dislocating (part of the force that acts parallel to the bone and away from the axis of rotation) component increases. Thus, the greatest torque is generated by a muscle when the direction of pull is at a 90° angle

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to the bone. The second influencing factor is attributed to the length-tension relationship of a muscle. The elbow flexors are a biarticulate muscle in that it crosses two articulations (elbow and shoulder). When the flexors contract, they tends to pull the two bony attachments (upper arm and lower arm) toward the center of the muscle and therefore tend to cause motion in the articulations that it crosses. When the shoulder joint is at a position of extension, the tension that is producible in the muscle progressively decreases as the muscle shortens to flex the elbow. Because the muscle is a two-joint muscle, certain adjustments to the muscle may be made as the muscle is shortened. As the elbow is flexed, the shoulder may flex, thus maintaining the length of the muscle. Consequently, shoulder flexion in conjunction with elbow flexion prevents the elbow flexors muscle from losing the tension advantage of its resting length in contrast to only elbow flexion. The ability of a two-joint muscle to maintain its length during contraction may allow the muscle greater tension when its attachment attains a greater than 90° angle to the bone (dislocating component). Based upon the observed movement at the shoulder joint and the application of the biomechanics of movement, the progressive increases in shoulder movement with corresponding increases in resistance are justified.

Trunk Angle

Examination of trunk angle mean values for each of the five conditions throughout the execution of the biceps curl also revealed a similar pattern of movement. In Phase I the trunk flexed, as indicated by the increase in trunk angle values. This type of movement is purely a compensatory adjustment made by the body, in an attempt to overcome the inertia of the stationary resistance the subjects flexed forward at the trunk. This act provided a burst of momentum that allowed movement to begin. In addition to overcoming the inertia of the stationary resistance,

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this movement also compensated for muscular deficiencies. This is illustrated in the magnitude of movement during this first phase. As the resistance was increased so did the magnitude of trunk flexion. Because the elbow flexors were not strong enough to initiate movement the trunk flexed forward to gain a mechanical advantage. As an added note, the starting position of the trunk was also influenced. In the heavier resistance conditions the trunk, at the initiation of movement, started in a more flexed position.

In Phase II the direction of trunk movement changed and the trunk began to extend. This movement was also a compensatory adjustment made by the body. Having flexed forward to overcome the inertia of the resistance, the trunk now extended to assist in movement. Again, this adjustment helped to maintain movement and compensate for lack of muscular strength. The magnitude of extension was more pronounced in the heavier resistance conditions. This pattern of extension persisted in the remaining phases of the biceps curl, except for Phase IV in the 100% RM condition. In Phase IV of the 100% RM condition the trunk changed direction and flexed. This movement was probably an attempt by the trunk to slow momentum, thus stabilizing the body and maintaining balance. Similar to the starting angles of the trunk, the trunk angles at the completion of movement were more extreme in the heavier resistance conditions. The observed type of trunk movement is consistent with that found in strength training literature, especially when heavier resistances are utilized.

Body Angle

Examination of body angle mean values for each of the five conditions revealed a similar movement pattern throughout certain phases of the biceps curl. In Phase I, similar to the trunk angle, the body angle demonstrated flexion. The causes of this movement are the same as that of the trunk. The body flexed in an attempt to

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overcome the inertia of the stationary resistance and compensate for muscular deficiencies. The magnitude of body flexion was directly related to resistance. The heavier resistance conditions caused a greater change in the body angle.

In Phase II the direction of body movement changed, with the body having extended. Similar to the trunk, the body extended to assist in movement. Again, the magnitude of extension was more pronounced in the heavier resistance conditions.

In Phase III the pattern of body angle extension persisted in the 90% RM and 100% RM conditions, while the body angle of the 60% RM, 70% RM, and 80% RM conditions remained comparable to the values at the end of Phase II. The absence of movement in these three conditions was most likely an attempt to stabilize the body. The continued extension of the body angle in the 90% RM and 100% RM conditions was again, an attempt to assist in movement.

In Phase IV changes in the body angle were quite diverse. In the 60% RM and 80% RM conditions the body angle remained unchanged, again suggesting that this area was serving a stabilitory role. In the 70% RM, 90% RM, and 100% RM conditions the direction of movement for the body angle changed and flexion was observed. This type of movement was an attempt to assist in the slowing of momentum and attainment of a state of balance. Strength training literature has not focused on this particular area of the body, and therefore the conclusions based upon the observed movement are solely those of the author.

Knee Angle

Examination of knee angle mean values for each of the five conditions revealed a similar type of movement pattern throughout the execution of the biceps curl. The extension of the knee observed in Phase I was, similar to the other joints, an attempt to help the body overcome the inertia of the stationary resistance. In Phase

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II the knee joint continued to extend, which indicated that this joint was still assisting in the motion of the curl. In Phase III the knee joint flexed, which suggested that the knee was serving a stabilitory role by absorbing the momentum gained throughout the previous phases. In Phase IV the knee joint extended, which was an attempt to slow momentum and stabilize the body upon the completion of the movement. Similar to the body angle, strength training literature has not focused on the knee joint during the execution of the biceps curl, therefore the observations are those of the author.

Related Factors

Several other factors may have influenced the magnitude of change observed in the selected angles. Muscular fatigue in relation to the order in which the conditions were performed may have affected the magnitude of movement in the selected angles. One would expect movement to become more pronounced in the heavy resistance conditions if they were performed in hierarchial order. If the order were switched and the conditions were performed in reverse the subject may have been fresh for the heavy resistance conditions and the magnitude of angular change may have been more overt in the lighter resistance conditions. Therefore, muscular fatigue in relation to condition order may have influenced the magnitude of change in joint angles. Motivation may also have affected performance. A subject may not have been as highly motivated in the later stages of testing as he was in the earlier stages, consequently, technique from one condition to another may have been affected. A learning effect may have occurred during the testing process. With only limited previous experience, the subject may have improved his own technique, regardless of the condition, throughout the testing process. Finally, the measured angles may have interacted to influence technique. Movement in one particular angle may have been more extensive, thus comprising change in the other angles.

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CHAPTER V

SUMMARY, FINDINGS, CONCLUSIONS, AND RECOMMENDATIONS

Summary

This study was undertaken to determine an optimal training intensity for the biceps curl that was biomechanically correct and would allow for optimal benefits in terms of physiological adaptations. The effect of increased resistance on biceps curl technique was investigated.

Ten college age males participated in the study. Prior to the testing process each subject's 1 RM was established in order to determine the resistances for the five conditions. The selected conditions for this study were 60%, 70%, 80%, 90%, and 100% of each individual's 1 RM.

All testing was completed in one afternoon. The testing procedure was identical for all 10 subjects, with the exception of the condition order. Each subject was required to perform one repetition for each of the five conditions. However, the order in which the conditions were performed was randomized to eliminate bias from the study. The subjects were filmed with a Photo-Sonic 1PL camera with their right side facing the camera. For this particular study only the concentric phase of each condition was filmed. At the completion of each condition the necessary weightload adjustments were made and the procedure was repeated until all five conditions were completed.

Data comparisons among subjects were made to determine the magnitude of differences with respect to body mechanics that occurred for each condition. Of

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particular interest were the changes that occurred in the shoulder, trunk, body, and knee joints.

Several statistical procedures were used to describe the data that was generated in the study. The mean of the group for each angle was calculated to compare the differences among the five conditions. In addition, F-ratios were also calculated to further describe the differences that occurred in the selected body angles among the five conditions.

Findings

The data generated from this study revealed the following significant findings:

1. Three distinguishable shoulder movement patterns were observed, with movement more overt in the 90% RM and 100% RM conditions.

2. Four different movement patterns were identified in the trunk, with the more movement present in the 100% RM condition.

3. Four different body movement patterns were observed, with movement more overt in the 100% RM condition.

4. There was an absence of patterned movement at the knee, however, movement was extensive in the 80% RM and 100% RM conditions.

5. F-ratio analysis indicated that the greatest amount of variance was present between the 60% RM and 100% RM conditions.

Conclusions

It was the belief of the researcher that varying intensities of an individual's 1 RM would produce biomechanical movements that are incorrect with respect to proper biceps curl technique. This was determined to be correct, with movement having become more improper as resistance was increased. In consideration of this, it was

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determined that lower intensities of an individual's 1 RM are more conducive to proper biceps curl technique.

The priorities of the individual will be reflected in the type of strength training program that is implemented, with regard to training intensity. Individuals who are using near maximum resistances in their strength training exercises may jeopardize their safety as well as possible physiological adaptation by compromising technique. Conversely, individuals who maintain proper technique by training at a lesser intensity may maximize physiological adaptation and reduce the likelihood of injury.

It is important to realize that strength training, regardless of intensity, will result in varied strength gains. A person who desires to improve muscular strength in the elbow flexors could implement any of the training intensities used in this study. The intensity that would be employed would vary based upon the experience and strength level of the individual. Proper technique is the key factor. The intensity that is conducive to proper technique is therefore individualized.

In conclusion, training at intensities considerably less than an individual's 1 RM can be conducive to proper technique and still allow for optimal strength gains.

Recommendations

Several statements can be made with respect to the data generated in this study. Higher training intensities did produce more movement in the measured angles. However, training intensity may not be the only factor that influenced technique. Variables such as fatigue and condition order, not investigated in this study, may have affected the magnitude of change in the measured angles.

Future studies in this area may want to implement some of the following:

1. Perform the selected conditions in a randomized order, however, perform multiple repetitions for each of the conditions. By doing this, the effect that fatigue in

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relation to condition order had on technique could be investigated.

2. Perform the sequenced conditions in a variety of orders over a period of several days. This will allow for a more detailed analysis of the effect that varied intensities have on technique.

3. Classify the subjects into categories with regard to strength training experience. This will allow the results to be generalized to only that specific population.

4. Film and analyze the biomechanical changes that occur during the eccentric phase of the biceps curl.

5. Employ more subjects for the study. A larger sample provides for more accurate generalizations to the general population.

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Appendix A

Human Subject Review Board Acceptance Form

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Human Subjects Institutional Review Board

Kalamazoo, Michigan 49008-3899

WESTERN MICHIGAN UNIVERSITY

Date: February 18, 1991

To: Jeffrey Gailhouse

From: Mary Anne Bunda, Chair Mary Anne Bunda

Re: HSIRB Project Number 91-01-27

This letter will serve as confirmation that your research protocol, "The Effect of Varying Weightload Intensities on Bicep Curl Technique," has been approved after expedited review by the HSIRB. The conditions and duration of this approval are specified in the Policies of Western Michigan University. You may now begin to implement the research as described in the approval application.

You must seek reapproval for any change in this design. You must also seek reapproval if the project extends beyond the termination date.

The Board wishes you success in the pursuit of your research goals.

xc: Mary Dawson, HPER

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Appendix B

Informed Consent

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APPENDIX B

Informed Consent

Jeff Gailhouse, a graduate student in the Department of Health, Physical Education and Recreation at Western Michigan University, is conducting research on the effect of varying weightload intensities on biceps curl technique. The purpose of the study is to investigate anatomical deviations from proper biceps curl technique that occur due to the intensity (weight) of the exercise. Research of this nature has not been performed previously and therefore, the results may be subject to further investigation. Your role in the project will be to attend a single one-hour session during which data collection will be performed by high speed filming of the full body. The data will be analyzed for the purpose of determining optimal training intensities for the biceps curl.

At this experimental session, you will perform the biceps curl. After determining the maximum amount of weight that you can curl, you will perform one repetition at 60%, 70%, 80%, 90%, and 100% of your maximum weight. There is a slight risk that you may experience temporary muscle soreness as a result of participation. While no direct benefit is likely to be experienced by yourself as a result of participation, your results may make it possible to select a training intensity that is mechanically correct yet yields optimal physiological adaptations.

Your name will be omitted from the published research findings. The film will remain in the possession of the biomechanics laboratory and will not be distributed for any reason. Upon completion of the project, the film will be destroyed. You reserve the right to terminate your participation during any phase of the project without consequence to your grades or your relationship with Western Michigan University. Should questions arise after this signing, you may contact Jeffrey Gailhouse at 624-1960.

I, _____, have read this statement and have had all my questions answered.

Date:_____

Signature:_____

Witness:_____

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Appendix C

Subject-Treatment Order

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APPENDIX C

Subject-Treatment Order

SUBJECT	1 RM	60%	70%	80%	90%	100%
1	105	65(1)	75(5)	85(2)	95(3)	105(4)
2	120	70(4)	85(2)	95(1)	110(5)	120(3)
3	85	50(2)	60(5)	70(4)	75(1)	85(3)
4	120	70(3)	85(5)	95(1)	110(4)	120(2)
5	110	65(3)	75(2)	90(1)	100(4)	110(5)
6	115	70(4)	80(2)	90(5)	105(1)	115(3)
7	105	65(1)	75(2)	85(5)	95(3)	105(4)
8	100	60(3)	70(1)	80(2)	90(4)	100(5)
9	100	60(3)	70(5)	80(1)	90(2)	100(4)
10	135	80(3)	95(5)	110(4)	120(1)	135(2)

Note. ( ) indicates order in which treatment was performed

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Appendix D

Anatomical Landmarks for Joint Angle Calculation

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APPENDIX D

Anatomical Landmarks for Joint Angle Calculation

Angle	Endpoint(P1)	Vertex(V)	Endpoint(P2)
Elbow	Right Wrist	Right Elbow	Right Shoulder
Shoulder	Right Elbow	Right Shoulder	Horizontal
Trunk	Right Shoulder	Right Hip	Vertical
Body	Right Shoulder	Right Ankle	Vertical
Knee	Right Hip	Right Knee	Right Ankle

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Appendix E

Subject Raw Angle Data

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APPENDIX E

Subject Raw Angle Data

Subject 1

Position

Condition 1	Initiation	135°	90°	45°	Completion
Frame No.	3	18	41	85	118
Elbow	167.6°	135.5°	90.6°	45.9°	33.0°
Shoulder	84.7°	94.4°	97.9°	79.9°	65.2°
Trunk	0.9°	0.6°	1.1°	-2.8°	-5.4°
Body	1.6°	1.3°	1.7°	0.3°	0.5°
Knee	176.6°	179.3°	177.2°	178.1°	178.6°

Condition 2	Initiation	135°	90°	45°	Completion
Frame No.	3	15	35	67	107
Elbow	165.2°	135.8°	90.0°	45.2°	23.9°
Shoulder	88.7°	97.1°	92.1°	68.5°	55.5°
Trunk	-0.5°	1.2°	0.4°	-2.7°	-5.6°
Body	3.0°	2.8°	1.4°	0.8°	1.6°
Knee	170.3°	172.2°	175.8°	176.1°	178.3°

Condition 3	Initiation	135°	90°	45°	Completion
Frame No.	3	28	54	110	124
Elbow	175.6°	134.4°	90.5°	45.0°	37.2°
Shoulder	84.0°	99.1°	99.1°	68.3°	53.2°
Trunk	2.0°	2.3°	-1.2°	-3.0°	-5.6°
Body	3.5°	3.7°	2.1°	0.7°	0.0°
Knee	173.6°	173.0°	173.3°	171.5°	172.2°

Condition 4	Initiation	135°	90°	45°	Completion
Frame No.	3	25	65	134	164
Elbow	174.3°	134.4°	90.7°	45.1°	30.9°
Shoulder	82.5°	100.4°	92.9°	59.1°	47.8°
Trunk	0.3°	2.6°	-0.6°	-2.9°	-6.5°
Body	1.8°	2.2°	-0.7°	0.5°	0.3°
Knee	171.8°	175.7°	179.0°	179.6°	180.5°

Condition 5	Initiation	135°	90°	45°	Completion
Frame No.	3	15	43	158	187
Elbow	162.7°	135.3°	89.6°	45.0°	27.0°
Shoulder	90.1°	103.1°	101.9°	58.2°	47.4°
Trunk	2.9°	6.6°	3.2°	-4.3°	-6.9°
Body	3.0°	4.0°	1.6°	0.0°	0.5°
Knee	174.4°	175.6°	183.2°	179.0°	181.9°

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Appendix E--Continued

Subject 2

	Position				
Condition 1	Initiation	135°	90°	45°	Completion
Frame No.	3	10	30	53	71
Elbow	146.6°	135.8°	89.0°	44.1°	18.7°
Shoulder	99.0°	100.2°	94.9°	72.0°	63.3°
Trunk	0.0°	0.5°	0.3°	-4.7°	-5.6°
Body	1.2°	1.0°	0.6°	-0.5°	0.0°
Knee	170.6°	169.9°	173.9°	169.9°	169.4°
Condition 2	Initiation	135°	90°	45°	Completion
Frame No.	3	7	32	53	60
Elbow	137.8°	135.2°	90.4°	44.8°	30.0°
Shoulder	107.9°	106.9°	83.5°	67.9°	65.4°
Trunk	5.0°	4.7°	-2.1°	-4.5°	-3.6°
Body	2.3°	2.2°	-1.0°	-0.6°	0.3°
Knee	171.9°	170.8°	171.4°	169.7°	170.6°
Condition 3	Initiation	135°	90°	45°	Completion
Frame No.	3	3	30	65	88
Elbow	132.7°	132.7°	90.0°	45.9°	22.9°
Shoulder	107.2°	107.2°	95.2°	71.6°	59.2°
Trunk	1.9°	1.9°	0.0°	-4.7°	-4.1°
Body	0.9°	0.9°	-0.4°	-0.5°	0.8°
Knee	165.7°	165.7°	168.0°	166.0°	164.3°
Condition 4	Initiation	135°	90°	45°	Completion
Frame No.	3	10	34	77	98
Elbow	142.8°	134.9°	90.1°	44.7°	27.2°
Shoulder	111.3°	108.9°	89.6°	62.0°	53.4°
Trunk	8.9°	6.9°	-4.8°	-9.0°	-6.0°
Body	3.7°	3.0°	-0.9°	-0.3°	1.5°
Knee	175.1°	175.3°	172.1°	168.1°	170.5°
Condition 5	Initiation	135°	90°	45°	Completion
Frame No.	3	3	33	102	113
Elbow	131.2°	131.2°	90.2°	45.1°	33.6°
Shoulder	115.5°	115.5°	88.8°	58.5°	55.6°
Trunk	9.9°	9.9°	-7.8°	-9.6°	-6.2°
Body	3.8°	3.8°	-2.0°	2.0°	3.4°
Knee	173.1°	173.1°	167.4°	163.4°	164.0°

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Appendix E--Continued

Subject 3

	Position				
Condition 1	Initiation	135°	90°	45°	Completion
Frame No.	3	17	32	52	86
Elbow	165.7°	134.9°	90.7°	45.0°	14.4°
Shoulder	90.0°	98.1°	94.2°	86.4°	78.9°
Trunk	-0.4°	-0.7°	-1.5°	-1.4°	-6.0°
Body	0.9°	0.3°	-0.3°	0.5°	0.0°
Knee	180.0°	181.3°	181.6°	180.3°	178.8°
Condition 2	Initiation	135°	90°	45°	Completion
Frame No.	3	14	29	43	70
Elbow	159.6°	136.0°	89.5°	45.4°	10.7°
Shoulder	98.6°	104.4°	96.9°	86.6°	71.2°
Trunk	2.0°	3.2°	-2.0°	-5.9°	-12.4°
Body	3.2°	3.2°	0.6°	0.0°	-1.2°
Knee	179.4°	181.9°	182.2°	180.5°	183.1°
Condition 3	Initiation	135°	90°	45°	Completion
Frame No.	3	18	33	47	78
Elbow	168.5°	134.4°	90.6°	45.9°	8.5°
Shoulder	91.8°	110.7°	99.9°	75.3°	54.9°
Trunk	-1.3°	8.0°	-5.5°	-19.7°	-14.4°
Body	1.3°	3.9°	0.0°	-3.0°	-1.3°
Knee	176.8°	178.5°	175.5°	171.2°	174.4°
Condition 4	Initiation	135°	90°	45°	Completion
Frame No.	3	19	50	75	106
Elbow	160.4°	135.7°	90.5°	44.3°	20.1°
Shoulder	89.8°	102.2°	91.4°	68.5°	60.8°
Trunk	-0.5°	1.2°	-7.7°	-17.2°	-16.1°
Body	0.9°	0.8°	-2.1°	-3.7°	-3.6°
Knee	180.3°	180.0°	181.5°	180.0°	178.2°
Condition 5	Initiation	135°	90°	45°	Completion
Frame No.	3	20	39	59	136
Elbow	168.9°	135.4°	90.2°	44.5°	14.5°
Shoulder	92.1°	109.8°	95.9°	65.3°	59.7°
Trunk	1.5°	7.7°	-12.1°	-41.6°	-11.5°
Body	1.6°	3.0°	-3.1°	-9.1°	-1.6°
Knee	180.0°	181.2°	177.2°	153.6°	178.7°

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Appendix E--Continued

Subject 4

	Position				
Condition 1	Initiation	135°	90°	45°	Completion
Frame No.	3	12	30	50	76
Elbow	160.3°	133.9°	89.0°	44.7°	14.5°
Shoulder	95.9°	104.7°	95.9°	79.1°	75.1°
Trunk	0.7°	1.1°	-0.8°	-2.5°	-4.0°
Body	1.6°	1.9°	-0.6°	-0.7°	-0.3°
Knee	175.9°	174.9°	174.5°	176.8°	175.4°
Condition 2	Initiation	135°	90°	45°	Completion
Frame No.	3	15	30	54	90
Elbow	160.6°	133.4°	89.4°	44.9°	11.4°
Shoulder	97.5°	105.8°	98.3°	83.6°	73.0°
Trunk	1.6°	3.1°	-0.9°	-2.6°	-8.9°
Body	1.1°	1.4°	-1.1°	-0.9°	-1.5°
Knee	164.8°	162.7°	166.7°	166.9°	172.5°
Condition 3	Initiation	135°	90°	45°	Completion
Frame No.	3	19	39	63	98
Elbow	166.8°	134.0°	90.3°	45.3°	9.8°
Shoulder	90.8°	104.1°	91.0°	76.0°	68.1°
Trunk	-2.6°	2.0°	-5.9°	-6.7°	-7.7°
Body	-0.8°	0.0°	-3.5°	-3.3°	-2.0°
Knee	170.2°	172.5°	176.3°	176.6°	177.3°
Condition 4	Initiation	135°	90°	45°	Completion
Frame No.	3	23	40	77	105
Elbow	167.8°	134.7°	89.3°	44.9°	12.9°
Shoulder	91.9°	113.8°	103.9°	71.2°	58.6°
Trunk	-0.4°	10.2°	0.6°	-10.4°	-12.4°
Body	0.0°	3.4°	-1.0°	-3.2°	-1.6°
Knee	164.7°	168.1°	171.3°	171.6°	173.7°
Condition 5	Initiation	135°	90°	45°	Completion
Frame No.	3	27	50	126	168
Elbow	168.3°	134.7°	89.9°	45.0°	12.8°
Shoulder	93.7°	113.2°	102.0°	73.5°	65.9°
Trunk	0.0°	9.9°	-2.6°	-20.3°	-10.8°
Body	0.3°	3.7°	-1.4°	-4.3°	1.6°
Knee	166.2°	165.9°	170.8°	168.4°	174.1°

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Appendix E--Continued

Subject 5

	Position				
Condition 1	Initiation	135°	90°	45°	Completion
Frame No.	3	19	35	52	68
Elbow	165.4°	134.6°	88.9°	44.4°	20.4°
Shoulder	99.8°	104.4°	98.4°	76.9°	67.7°
Trunk	0.4°	0.0°	-4.4°	-6.9°	-7.8°
Body	1.6°	1.4°	-0.4°	-1.2°	-0.7°
Knee	181.5°	179.4°	180.6°	181.1°	178.7°
Condition 2	Initiation	135°	90°	45°	Completion
Frame No.	3	16	32	50	73
Elbow	157.2°	135.7°	89.6°	44.5°	10.4°
Shoulder	106.7°	111.9°	106.0°	84.7°	71.9°
Trunk	1.4°	3.5°	0.7°	-6.8°	-5.4°
Body	2.1°	2.7°	1.2°	-0.7°	1.5°
Knee	178.9°	181.1°	180.0°	180.3°	178.0°
Condition 3	Initiation	135°	90°	45°	Completion
Frame No.	3	20	37	67	98
Elbow	165.2°	134.8°	90.7°	45.8°	11.1°
Shoulder	103.0°	113.5°	101.3°	70.4°	66.5°
Trunk	2.2°	6.9°	-1.6°	-7.1°	-5.6°
Body	3.0°	4.2°	0.7°	-0.4°	2.0°
Knee	179.2°	180.0°	181.3°	179.3°	175.5°
Condition 4	Initiation	135°	90°	45°	Completion
Frame No.	3	24	44	95	128
Elbow	164.5°	134.2°	89.5°	45.0°	16.0°
Shoulder	101.6°	112.0°	93.7°	70.6°	61.3°
Trunk	1.9°	5.6°	-8.8°	-13.6°	-7.2°
Body	2.6°	3.8°	-1.3°	-1.8°	1.7°
Knee	181.3°	180.9°	180.0°	178.4°	180.6°
Condition 5	Initiation	135°	90°	45°	Completion
Frame No.	3	22	38	90	108
Elbow	143.0°	134.7°	91.1°	45.2°	19.3°
Shoulder	116.7°	125.1°	94.9°	58.5°	55.4°
Trunk	2.3°	15.8°	-9.0°	-17.3°	-11.8°
Body	2.4°	6.7°	-2.5°	-2.9°	0.0°
Knee	178.0°	169.9°	174.7°	177.1°	179.1°

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Appendix E--Continued

Subject 6

Position

Condition 1	Initiation	135°	90°	45°	Completion
Frame No.	3	6	35	53	63
Elbow	137.2°	134.8°	90.2°	44.9°	29.0°
Shoulder	105.6°	106.4°	73.0°	61.9°	59.2°
Trunk	2.1°	2.3°	-3.5°	-2.2°	-1.6°
Body	1.2°	1.3°	-1.4°	1.0°	1.7°
Knee	177.8°	178.9°	178.8°	178.5°	176.8°

Condition 2	Initiation	135°	90°	45°	Completion
Frame No.	3	15	40	61	68
Elbow	142.3°	135.3°	90.2°	45.1°	39.9°
Shoulder	101.0°	116.4°	77.7°	62.4°	59.8°
Trunk	2.0°	13.7°	-1.2°	-4.7°	-4.9°
Body	2.7°	6.1°	-0.6°	0.9°	1.2°
Knee	173.8°	175.5°	178.6°	179.0°	179.1°

Condition 3	Initiation	135°	90°	45°	Completion
Frame No.	3	21	42	57	69
Elbow	144.4°	134.8°	90.8°	44.2°	25.4°
Shoulder	103.4°	116.1°	67.6°	54.1°	51.8°
Trunk	1.1°	13.8°	-8.3°	-7.1°	-6.6°
Body	1.6°	5.8°	-2.6°	-1.0°	0.5°
Knee	174.5°	176.3°	178.7°	178.2°	177.8°

Condition 4	Initiation	135°	90°	45°	Completion
Frame No.	3	32	54	82	93
Elbow	156.2°	135.5°	89.3°	44.3°	32.6°
Shoulder	105.5°	90.9°	59.4°	52.9°	50.8°
Trunk	3.3°	3.0°	-16.5°	-13.0°	-10.6°
Body	0.0°	-0.8°	-6.4°	-4.2°	-2.6°
Knee	168.1°	171.4°	170.1°	172.9°	171.4°

Condition 5	Initiation	135°	90°	45°	Completion
Frame No.	3	22	51	81	93
Elbow	141.3°	134.3°	89.5°	45.4°	30.9°
Shoulder	110.0°	126.2°	64.8°	54.5°	53.1°
Trunk	4.9°	20.3°	-17.6°	-15.2°	-11.6°
Body	2.1°	7.0°	-5.6°	-3.0°	-1.0°
Knee	175.1°	180.4°	178.8°	180.0°	179.1°

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Appendix E--Continued

Subject 7

Position

Condition 1	Initiation	135°	90°	45°	Completion
Frame No.	3	12	25	37	49
Elbow	161.4°	135.7°	90.6°	44.1°	22.6°
Shoulder	92.8°	100.4°	100.8°	95.9°	84.1°
Trunk	-1.3°	0.7°	0.4°	1.0°	-3.1°
Body	1.0°	1.2°	0.3°	1.0°	0.4°
Knee	177.5°	180.2°	178.5°	174.3°	174.3°

Condition 2	Initiation	135°	90°	45°	Completion
Frame No.	3	12	27	42	63
Elbow	159.1°	136.2°	89.1°	46.0°	12.7°
Shoulder	97.9°	105.7°	101.3°	85.6°	62.2°
Trunk	-1.0°	3.6°	0.9°	-1.8°	-6.8°
Body	1.1°	1.6°	0.0°	-0.9°	-0.9°
Knee	179.2°	180.2°	178.5°	179.4°	175.7°

Condition 3	Initiation	135°	90°	45°	Completion
Frame No.	3	8	21	45	68
Elbow	147.9°	134.1°	90.3°	45.1°	21.3°
Shoulder	100.6°	107.4°	114.6°	78.2°	64.5°
Trunk	2.7°	5.2°	2.1°	-6.0°	-6.0°
Body	2.5°	3.1°	1.3°	-2.6°	-1.2°
Knee	179.4°	178.1°	179.2°	176.2°	174.7°

Condition 4	Initiation	135°	90°	45°	Completion
Frame No.	3	13	31	58	73
Elbow	162.3°	135.2°	90.7°	45.2°	21.9°
Shoulder	93.8°	106.3°	105.4°	69.4°	58.1°
Trunk	-2.7°	3.1°	-4.2°	-11.0°	-10.4°
Body	0.0°	1.1°	-1.6°	-3.9°	-2.5°
Knee	177.9°	178.2°	175.6°	174.3°	173.8°

Condition 5	Initiation	135°	90°	45°	Completion
Frame No.	3	8	28	54	83
Elbow	143.7°	135.4°	90.4°	45.2°	13.7°
Shoulder	105.0°	108.6°	105.7°	75.8°	55.1°
Trunk	5.5°	7.1°	-8.3°	-24.7°	-13.3°
Body	2.9°	3.1°	-2.0°	-6.4°	-2.3°
Knee	177.8°	178.7°	178.3°	178.5°	172.8°

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Appendix E–Continued

Subject 8

Position					
Condition 1	Initiation	135°	90°	45°	Completion
Frame No.	3	3	20	36	48
Elbow	123.6°	123.6°	91.7°	44.2°	21.1°
Shoulder	95.9°	95.9°	83.0°	67.9°	62.4°
Trunk	-4.1°	-4.1°	-6.1°	-7.6°	-10.1°
Body	0.5°	0.5°	-1.1°	-0.8°	-0.6°
Knee	178.3°	178.3°	180.0°	179.3°	176.3°
Condition 2	Initiation	135°	90°	45°	Completion
Frame No.	3	17	35	50	60
Elbow	145.6°	134.7°	88.9°	46.5°	27.3°
Shoulder	88.1°	85.8°	76.8°	68.0°	63.3°
Trunk	-4.6°	-4.8°	-7.4°	-6.5°	-8.8°
Body	-0.9°	-1.6°	-2.8°	-0.8°	-0.4°
Knee	179.7°	184.7°	183.5°	180.0°	179.2°
Condition 3	Initiation	135°	90°	45°	Completion
Frame No.	3	3	35	51	60
Elbow	130.6°	130.6°	91.1°	44.0°	21.8°
Shoulder	96.2°	96.2°	76.1°	66.4°	60.6°
Trunk	-1.7°	-1.7°	-6.9°	-7.9°	-9.7°
Body	1.2°	1.2°	-1.5°	-0.4°	0.5°
Knee	183.3°	183.3°	182.3°	179.1°	179.1°
Condition 4	Initiation	135°	90°	45°	Completion
Frame No.	3	3	35	54	68
Elbow	119.7°	119.7°	90.1°	46.3°	24.3°
Shoulder	98.3°	98.3°	71.3°	60.5°	57.0°
Trunk	-4.4°	-4.4°	-13.4°	-11.1°	-12.0°
Body	-0.5°	-0.5°	-4.6°	-1.4°	-0.4°
Knee	184.7°	184.7°	177.7°	179.7°	180.0°
Condition 5	Initiation	135°	90°	45°	Completion
Frame No.	3	3	29	47	58
Elbow	111.1°	111.1°	89.7°	44.4°	24.1°
Shoulder	113.1°	113.1°	74.6°	60.9°	54.3°
Trunk	1.9°	1.9°	-16.1°	-15.2°	-13.8°
Body	2.4°	2.4°	-4.8°	-2.6°	-0.4°
Knee	182.5°	182.5°	182.1°	178.6°	180.0°

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Appendix E–Continued

Subject 9

Position					
Condition 1	Initiation	135°	90°	45°	Completion
Frame No.	3	12	25	40	73
Elbow	156.5°	135.4°	89.3°	45.4°	17.8°
Shoulder	97.4°	103.7°	105.3°	100.8°	73.3°
Trunk	-0.4°	3.4°	1.5°	-0.4°	-7.6°
Body	0.4°	0.8°	-0.3°	-1.1°	-2.6°
Knee	166.4°	164.3°	168.0°	170.2°	169.0°
Condition 2	Initiation	135°	90°	45°	Completion
Frame No.	3	7	24	48	78
Elbow	139.9°	135.5°	91.6°	44.7°	22.2°
Shoulder	107.1°	107.3°	104.0°	84.0°	77.1°
Trunk	1.5°	2.4°	0.8°	-5.4°	-7.8°
Body	-0.3°	0.0°	-1.2°	-3.2°	-3.5°
Knee	165.0°	165.8°	171.3°	178.6°	174.8°
Condition 3	Initiation	135°	90°	45°	Completion
Frame No.	3	17	33	55	93
Elbow	159.4°	136.2°	90.1°	45.7°	19.4°
Shoulder	100.6°	107.1°	104.3°	89.9°	62.3°
Trunk	0.4°	4.7°	1.8°	-2.2°	-10.3°
Body	0.0°	0.6°	-0.9°	-2.5°	-4.5°
Knee	167.4°	168.9°	172.0°	173.3°	174.9°
Condition 4	Initiation	135°	90°	45°	Completion
Frame No.	3	16	27	50	108
Elbow	160.1°	134.7°	91.1°	45.1°	33.6°
Shoulder	105.2°	113.6°	117.6°	98.0°	72.0°
Trunk	7.7°	9.4°	7.7°	2.2°	-6.4°
Body	3.6°	3.9°	3.2°	1.2°	-2.7°
Knee	160.0°	161.0°	167.7°	173.5°	178.1°
Condition 5	Initiation	135°	90°	45°	Completion
Frame No.	3	12	29	188	198
Elbow	154.1°	135.5°	90.6°	45.1°	27.7°
Shoulder	107.9°	115.3°	115.8°	59.6°	53.8°
Trunk	5.0°	6.8°	6.0°	-16.2°	-13.3°
Body	0.6°	1.2°	0.8°	-5.6°	-4.1°
Knee	161.4°	158.5°	171.5°	178.0°	177.8°

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Appendix E–Continued

Subject 10

	Position				
Condition 1	Initiation	135°	90°	45°	Completion
Frame No.	3	6	32	85	93
Elbow	142.3°	134.4°	90.6°	44.9°	35.7°
Shoulder	107.0°	109.1°	84.2°	67.2°	65.1°
Trunk	4.6°	5.6°	-4.8°	-9.3°	-8.9°
Body	3.5°	3.9°	-1.5°	-0.5°	0.0°
Knee	181.5°	181.6°	181.8°	181.2°	180.4°
Condition 2	Initiation	135°	90°	45°	Completion
Frame No.	3	19	48	86	95
Elbow	167.8°	136.4°	89.3°	45.4°	34.0°
Shoulder	93.6°	112.4°	75.4°	62.6°	58.1°
Trunk	1.7°	9.6°	-7.1°	-9.7°	-9.5°
Body	2.1°	4.8°	-2.9°	-1.3°	0.0°
Knee	181.9°	180.5°	179.4°	177.6°	177.9°
Condition 3	Initiation	135°	90°	45°	Completion
Frame No.	3	10	31	100	110
Elbow	150.9°	136.0°	90.6°	45.4°	37.6°
Shoulder	108.0°	116.5°	97.0°	60.4°	58.5°
Trunk	5.0°	9.0°	-5.1°	-11.2°	-9.3°
Body	3.1°	4.8°	-1.6°	-0.5°	0.5°
Knee	179.0°	178.4°	179.4°	178.0°	177.5°
Condition 4	Initiation	135°	90°	45°	Completion
Frame No.	3	17	38	89	99
Elbow	165.5°	134.3°	89.4°	44.9°	34.7°
Shoulder	92.1°	116.9°	100.7°	67.2°	61.9°
Trunk	-1.8°	8.5°	-3.3°	-10.7°	-10.3°
Body	1.3°	5.4°	-1.0°	-1.0°	0.0°
Knee	177.6°	180.0°	180.3°	180.8°	181.3°
Condition 5	Initiation	135°	90°	45°	Completion
Frame No.	3	13	37	169	173
Elbow	156.3°	133.9°	90.3°	44.8°	37.8°
Shoulder	103.9°	119.0°	92.6°	61.1°	63.3°
Trunk	3.7°	10.8°	-10.4°	-17.7°	-15.8°
Body	2.8°	5.2°	-3.2°	-3.7°	-3.1°
Knee	181.1°	181.0°	176.0°	177.9°	177.3°

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Appendix F

Group Means by Joint Angle

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APPENDIX F

Group Means by Joint Angle

Shoulder

Position	Condition				
	1	2	3	4	5
Initiation	96.8°	98.7°	98.6°	97.2°	104.8°
135°	101.7°	105.4°	107.8°	106.3°	103.9°
90°	92.8°	91.2°	94.6°	92.6°	93.7°
45°	78.8°	75.4°	71.1°	67.9°	62.6°
Completion	69.4°	65.8°	60.0°	58.2°	56.4°

Trunk

Position	Condition				
	1	2	3	4	5
Initiation	0.3°	0.9°	1.0°	1.2°	3.8°
135°	0.9°	4.0°	5.2°	4.6°	9.7°
90°	-1.8°	-2.5°	-3.1°	-5.1°	-7.5°
45°	-3.7°	-5.3°	-7.6°	-9.7°	-18.2°
Completion	-6.0°	-7.5°	-7.9°	-9.8°	-11.5°

Body

Position	Condition				
	1	2	3	4	5
Initiation	1.4°	1.4°	1.6°	1.3°	2.2°
135°	1.4°	2.3°	2.8°	2.2°	4.0°
90°	-0.1°	-0.6°	-0.6°	-1.6°	-2.2°
45°	-0.2°	-1.8°	-1.4°	-1.8°	-3.6°
Completion	-0.2°	0.2°	-0.5°	-1.0°	-0.7°

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Appendix F--Continued

Knee					
Position	Condition				
	1	2	3	4	5
Initiation	176.6°	174.5°	174.9°	174.2°	175.0°
135°	176.8°	175.3°	175.5°	175.5°	174.6°
90°	177.5°	176.8°	176.6°	175.5°	176.0°
45°	176.9°	176.5°	174.9°	175.9°	173.5°
Completion	175.8°	176.8°	174.8°	176.8°	176.5°

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